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Threats and Attacks on IOT Based Smart Campus System



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STATEMENT OF SUBMISSION

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CHAPTER 1

INTRODUCTION

Abstract --
The Internet of Things (IoT) is the most promising area which penetrates the advantages of Wireless Sensor and Actuator Networks (WSAN) and Pervasive Computing domains. Different applications of IoT have been developed and researchers of IoT well identified the opportunities, problems, challenges and the technology standards used in IoT such as Radio-Frequency Identification (RFID) tags, sensors, actuators, mobile phones, etc. This

paper is of two-fold; the first fold covers the different applications that adopted smart technologies so far. The second fold of this paper presents the overview of the sensors and its standards.

1. INTRODUCTION

Internet of Things (IoT) is a new revolution of the Internet. It makes Objects themselves recognizable, obtain intelligence, communicate information about themselves and they can access information that has been aggregated by other things. The Internet of Things allows people and things to be connected Anytime, Anyplace, with Anything and Anyone, ideally using Any path/network and Any service. This implies addressing elements such as Convergence, Content, Collections, Computing, Communication, and Connectivity.

The Internet of Things provides interaction among the real/physical and the digital/virtual worlds. The physical entities have digital counterparts and virtual representation and things become context aware and they can sense, communicate, interact, exchange data, information and knowledge. Through the use of intelligent decision-making algorithms in software applications, appropriate rapid responses can be given to physical entity based on the very latest information collected about physical entities and consideration of patterns in the historical data, either for the same entity or for similar entities. These paves new dimension of IoT concept in the domains such as supply chain management, transportation and logistics, aerospace, and automotive, smart environments (homes, buildings, infrastructure), energy, defense, agriculture, retail and more.

The vision of IoT is to use smart technologies to connect things any-time, any-place for anything. The IoT was started in the year 1998 and the term Internet of Things was first coined by Kevin Ashton in 1999.

The Internet of Things has been evolved in a tremendous way over the past decade and still IoT is an emerging trend for researchers in both academia and industry. Many findings of IoT reported in literature presents meaningful definitions. According to CASAGRAS project [1]: “A global network infrastructure linking physical and virtual objects through the exploitation of data capture and communication capabilities. This infrastructure includes existing and evolving Internet and network developments.

According to the IEEE Internet of Things journal, An IoT system is a network of networks where, typically, a massive number of objects/things/sensors/devices are connected through communications and information infrastructure to provide value-added services via intelligent data processing and management for different applications. The Internet of Things (IoT) is a computing concept that describes a future where every day physical objects will be connected to the Internet and will be able to identify themselves to other devices. The term is closely identified with RFID as the method of communication, although it could also include other sensor technologies, other wireless technologies, QR codes, etc. According to The Internet of Things European Research Cluster (IERC) definition [3] states that IoT is a dynamic global network

infrastructure with self-configuring capabilities based on standard and interoperable communication protocols where physical and virtual “things” have identities physical attributes, and virtual personalities and use intelligent interfaces, and are seamlessly integrated into the information network.

This paper presents the survey which gives a picture of the current state of the art on the IoT. More specifically, it provides clear insight to readers about the different visions of the Internet of Things paradigm and illustrates the benefits of this paradigm in everyday-life. This also provides the application domains of IoT and IT enabled communication technologies and standards used so far.

1.1. APPLICATION DOMAINS

The Applications of the IoT are numerous and diversified in all areas of everyday life of people which broadly covers society, industries, and environment. All the IoT applications developed so far comes under these three broad areas as shown in Table 1. According to Internet of Things Strategic Research Agenda (SRA) during 2010, 6 or more application domains were identified that are smart energy, smart health, smart buildings, smart transport, smart living and smart cities. According to the survey that the IoT-I project ran during 2010 65 IoT application scenarios were identified and grouped in to 14 domains, which are Transportation, Smart Home, Smart City, Lifestyle, Retail, Agriculture, Smart Factory, Supply chain, Emergency, Health care, User interaction, Culture and tourism, Environment and Energy. Some of the IoT applications are briefly explained in next coming paragraphs.

1.1.1. Society

Activities related to the betterment and development of society, cities and people Smart Cities, Smart Animal Farming, Smart Agriculture, Healthcare, Domestic and Home automation, Independent Living, Telecommunications, Energy, Defense, Medical technology, Ticketing, Smart Buildings.

1.1.2. Environment

Activities related to the protection, monitoring and development of all-natural resources Smart Environment, Smart Metering, Smart Water Recycling, Disaster Alerting

1.1.3. Industry

Activities related to financial, commercial transactions between companies, organizations and other entities Retail, Logistics, Supply Chain Management Automotive, Industrial Control, Aerospace and Aviation.

1.2. Role of IoT

The IoT play a vital role to improve the smartness of cities includes many applications to monitoring of parking spaces availability in the city, monitoring of vibrations and material conditions in buildings and bridges, sound monitoring in sensitive areas of cities, monitoring of vehicles and pedestrian levels, intelligent and weather adaptive lighting in street lights, detection of waste containers levels and trash collections, smart roads, intelligent highways with warning messages and diversions according to climate conditions and unexpected events like accidents or traffic jams. Some of IoT smart cities applications are smart parking, structural health, noise urban maps, traffic congestion, smart lightning, waste management, intelligent transportation systems and smart building. These smart cities IoT applications use RFID, Wireless Sensor Network and Single sensors as IoT elements and the bandwidth of these applications ranges from small to large.

The IoT can help to improve and strengthen the agriculture work by monitoring soil moisture and trunk diameter in vineyards to control and maintain the amount of vitamins in agricultural products, control micro climate conditions to maximize the production of fruits and vegetables and its quality, study of weather conditions in fields to forecast ice information, rail, drought, snow or wind changes, control of humidity and temperature level to prevent fungus and other microbial contaminants. The role of IoT in water management includes study of water suitability in rivers and the sea for agriculture and drinkable use, detection of liquid presence outside tanks and pressure variations along pipes and monitoring of water level variations in rivers, dams and reservoirs. This kind of IoT applications use Wireless sensor network and single sensors as IoT elements and the bandwidth range as medium.

Implementing the IoT in Retail/Supply Chain Management has many advantages which include monitoring of storage conditions along the supply chain and product tracking for traceability purposes and payment processing based on location or activity duration for public transport, gyms, theme park, etc. In the shop itself, IoT offers many applications like guidance in the shop according to a preselected shopping list, fast payment solutions like automatically check-out using biometrics, detection of potential allergen in a given product and control of rotation of products in shelves and warehouses to automate restocking processes. The IoT elements used in this kind of application are RFID and WSN and the bandwidth range is small.

1.2.1. Health Care:

Many benefits provided by the IoT technologies to the healthcare domain are classified into tracking of objects, staff and patients, identification and authentication of people, automatic data collection and sensing. Tracking is the function used to identify a person or an object in motion. This includes the case of patient flow monitoring to improve workflow in hospitals. The identification and authentication include patient identification to reduce incidents harmful to patients, comprehensive and current electronic medical record maintenance, and infant identification in hospitals to prevent mismatching. The automatic data collection and transfer is mostly aimed at reducing form processing time, process automation, automated care and procedure auditing, and

medical inventory management. Sensor devices enable function centered on patients, and in particular on diagnosing patient conditions, providing real-time information on patient health indicators. Application domains include different telemedicine solutions, monitoring patient compliance with medication regiment prescriptions, and alerting for patient well-being. In this capacity, sensors can be applied both in in-patient and out-patient care. The elements of IoT in Health Care are RFID, NFC, WSN, WIFI, Bluetooth, etc. significantly improve the measurement and monitoring methods of vital functions such as temperature, blood pressure, heart rate, cholesterol level, blood glucose, etc.

1.2.2. Security & Emergencies:

The IoT technologies in the field of security and emergencies are tremendously increased in which few are listed; perimeter access control, liquid presence, radiation levels and explosive and hazardous gases, etc. The perimeter access control is used to detect and control the unauthorized people entry to restricted areas. The liquid presence is used for liquid detection in data centers, warehouses and sensitive building grounds to prevent break downs and corrosion. The radiation levels application used to measure the radiation levels in nuclear power stations surroundings to generate leakage alerts and the final IoT application is used to detect the gas levels and leakages in industrial environments, surroundings of chemical factories and inside mines.

1.3. IoT COMMUNICATION TECHNOLOGIES

The communication enabling technologies of IoT heavily depends on rapid technical innovation in 4 fields; technology used to connect everyday objects and devices to large databases and networks, technology used for data collection with ability to detect changes in the physical status of objects, technology to take action through embedded intelligence in objects, and finally to make smaller and smaller things will have the ability to interact and connect. The combination of all these developments made the effective and efficient communications on IoT applications.

1.3.1. RFID:

RFID is not new and it was popular in the early 20th century. Initially, it was based on radio waves and later radio waves combined with radar signals. They can be used to provide P2P connection between objects. RFID consists of three main components such as a transponder or tag to carry data, which is located on the object to be identified, an interrogator or reader, which reads the transmitted data, and Middleware, which forward the data to another system, such as a database, a PC or robot control system. Frequencies currently used for data transmission by RFID typically include 125 kHz (low frequency), 13.56 MHz (high frequency), or 800-960 MHz (ultra-high frequency). RFID is set to revolutionize the retail sector. By 2008, according to

IDTechEx, retailers worldwide are expected to account for over USD 1.3 billion of a global RFID market of USD 7 billion. RFID standards relate both to frequency protocols (for data communication) and data format.

1.3.2. Sensors:

Sensors are one of the key building blocks of the Internet of Things which can be deployed everywhere from military battlefields to vineyards. A sensor is an electronic device, which detects senses or measures physical stimuli and responds to it in a specific way. It converts signals from stimuli into an analogue or digital form, so that the raw data about detected parameters are readable by machines and humans. Sensors can also be implanted under human skin, in a purse or on a dress. Some can be as small as four millimeters in size, but the data they collect can be received hundreds of miles away.
Sensors complement human senses and have become indispensable in a large number of industries, from health care to construction. Sensors have the key advantage that they can anticipate human needs based on information collected about their context.Common applications of sensors include military, environment, healthcare, construction, commercial applications, home applications, etc. When a sensor forms part of a sensor network, it is known as a sensor "node". While it is now easy to deploy single sensors, ensuring connectivity between multiple nodes is a more challenging task. Sensor nodes can be connected to each other in two ways: wire and wireless. A sensor node in a wireless sensor network is a small low power device with power-supply, data storage, microprocessors, low-power radio, analogue-to-digital converters (ADCs), data transceivers, and controllers. Wireless sensor networks offer solutions for a number of sectors, such as health care, security and agriculture.

1.3.3. RFID and Sensors:

The progressive combination of communication technologies and microelectronics gradually removes boundaries between physical objects and the virtual networked world. The main function of an RFID tag is to identify and track what, which and where the object accurately. Sensor technology provides information about the external environment and circumstances surrounding an object. The integration of wireless sensing technologies with RFID tags on moving objects provides a fuller picture about their location and status. The main distinguishing feature of an RFID sensor tag from a normal RFID tag is that, apart from tracking and monitoring functions, sensor-enabled RFID can take action on the basis of data collected by the sensor. These two technologies, in combination with modern wireless networks, create opportunities for a myriad of applications in national security, military field, agriculture, medicine, retail, food industry and many other sectors of the economy.

1.3.4. Sensors and Mobile Phones:

Mobile phones are already an integral part of everyday life for many people. Due to their widespread use, mobile networks play a key role in bringing new "ubiquitous"

communication technologies to the masses. Today, mobile phones are not only a device for making calls, but it equipped with data, text and video streaming functions. Currently, the combination of sensors with mobile phones offers several possible applications such as device for relaying data collected by sensors, touch sensors, movement recognition, sensing the status of their environment through smell sensors, etc.

1.3.5. Near Field Communication:

Near field communication (NFC) is a set of standards for smart phones and similar mobile devices to establish communication with each other by touching them together or bringing them together no more than a few inches. NFC devices can be used in contactless payment systems, similar to those currently used in credit cards and electronic ticket smartcards, and allow mobile payment to replace or supplement these systems. The mobile OS Android Beam uses NFC to complete the steps of enabling, pairing and establishing a Bluetooth connection when doing a file transfer

1.3.6. ZigBee:

ZigBee is a specification standard for a suite of high-level communication protocols used to create personal area networks built from small, low-power digital radios. ZigBee is based on an IEEE 802.15 standard. Though low-powered, ZigBee devices often transmit data over longer distances by passing data through intermediate devices to reach more distant ones, creating a mesh network. They can used in applications that require a low data rate, long battery life, and secure networking.

2. Feasibility report of IoT and its Security

From Nearly Past the main problem with the "Internet of Things" was the whole data architecture of the system. I don't want my thermostat talking directly to my refrigerator or my smoke alarms sending text messages directly to my phone. I don't want my lights to come on because my security alarm knows how to send direct commands to the light controller. I don't want all my sensor data uploaded to some Internet web site (encrypted or not) so that some company can gather statistics on what time I go to bed.

I want a data management layer that sits between all my smart devices and lets me control everything. I want that data layer to be able to exist on storage devices that I control, not some third party. The data layer would allow all the smart devices to create "smart data objects" that can store sensor information and to take specific actions if certain data objects exist. The only things that smart devices need to know is how to create these data objects and how to take specific actions when other kinds of data objects exist. Security and privacy are handled by this data management layer, not the devices themselves.

With such a system in place, the only thing my smart smoke detector needs to be able to do is create a "smoke detected" data object or a "current battery level" data object. It doesn't need to know how to take any kind of action beyond that. Now all I need is an application that I control that is monitoring the data storage layer for any "smoke detected" objects. This application will check my custom policy (that I created) and determine if it should create a "call fire department" data object. My smart phone (that knows how to call the fire department) watches for such objects and will call if it finds one. The smoke detector and the smart phone don't need to know anything about each other. If later I decide that I want to change my policy so that my garage door also opens and my smart car backs itself out of the garage when a "smoke detected" object appears, then neither the phone nor the smoke detector has to change a thing.

Now a days, The Internet of Things (IoT) has been an industry buzzword for years, but sluggish development and limited commercialization have led some industry watchers to start calling it the “Internet of NoThings”.

IoT development is in trouble. Aside from spawning geeky jokes unfit for most social occasions, the hype did not help and in fact, I believe it actually caused a lot more harm than good. There are a few problems with IoT, but all the positive coverage and baseless hype are one we could do without. The upside of generating more attention is clear: more investment, more VC funding, more consumer interest.

2.1. Security and the Internet of Things:

However, these come with an added level of scrutiny, which has made a number of shortcomings painfully obvious. After a couple of years of bullish forecasts and big promises, IoT security seems to be the biggest concern. The first few weeks of 2015 were not kind to this emerging industry, and most of the negative press revolved around security.

Was it justified? Was it just “fear, uncertainty and doubt” (FUD), brought about by years of hype? It was a bit of both; although some issues may have been overblown, the problems are very real, indeed.

From “Year Of IOT” To Annus Horribilis For IoT

Many commentators described 2015 as “the year of IoT,” but so far, it has been a year of bad press. Granted, there are still ten months to go, but negative reports keep piling on. Security firm Kaspersky recently ran a damning critique of IoT security challenges, with an unflattering headline, “Internet of Crappy Things”.

Kaspersky is no stranger to IoT criticism and controversy; the firm has been sounding alarm bells for a while, backing them up with examples of hacked smart homes, carwashes and even police surveillance systems. Whether a hacker wants to wash their ride free of charge, or stalk someone using their fitness tracker – IoT security flaws could make it possible.

Wind River published a white paper on IoT security in January 2015, and the report starts off with a sobering introduction. Titled Searching For The Silver Bullet, it summarizes the problem in just three paragraphs, which I will condense into a few points:

- Security must be the foundational enabler for IoT.
- There is currently no consensus on how to implement security in IoT on the device.
- A prevalent, and unrealistic, expectation is that it is somehow possible to compress 25 years of security evolution into novel IoT devices.
- There is no silver bullet that can effectively mitigate the threats.

However, there is some good news; the knowledge and experience are already here, but they have to be adapted to fit the unique constraints of IoT devices.
Unfortunately, this is where we as system security developers stumble upon another problem, a hardware problem.

U.S. Federal Trade Commission chairwoman, Edith Ramirez, addressed the Consumer Electronics Show in Las Vegas earlier this year, warning that embedding sensors into everyday devices, and letting them record what we do, could pose a massive security risk.

2.2. Ramirez outlined three key challenges for the future of IoT:

- Ubiquitous data collection.
- Potential for unexpected uses of consumer data.
- Heightened security risks.

She urged companies to enhance privacy and built secure IoT devices by adopting a security-focused approach, reducing the amount of data collected by IoT devices, and increasing transparency and providing consumers with a choice to opt-out of data collection.

Ramirez went on to say that developers of IoT devices have not spent time thinking about how to secure their devices and services from cyber-attacks.

"The small size and limited processing power of many connected devices could inhibit encryption and other robust security measures," said Ramirez. "Moreover, some connected devices are low-cost and essentially disposable. If a vulnerability is discovered on that type of device, it may be difficult to update the software or apply a patch – or even to get news of a fix to consumers."

While Ramirez is spot on in most respects, I should note that the Internet went through a similar phase two decades ago. There were a lot of security concerns, and the nineties saw the emergence of the internet-borne malware, DDoS attacks, sophisticated phishing and more. Even though Hollywood depicted a dystopian future in some films, we have ended up with kittens on social networks and a high-profile security breach here and there.

The Internet is still not secure, so we can't expect IoT to be secure, either. However, security is constantly evolving to meet new challenges, we've seen it before, and we'll see it again, with IoT and subsequent connected technologies.

IoT Hardware Is And Will Remain A Problem

Some of you will be thinking that the hardware issues mentioned by the FTC boss will be addressed yes, some of them probably will be addressed.

As the IoT market grows, we will see more investment, and as hardware matures, we will get improved security. Chipmakers like Intel and ARM will be keen to offer better security with each new generation, since security could be a market differentiator, allowing them to grab more design wins and gain a bigger share.

Technology always advances, so why not? New manufacturing processes generally result in faster and more efficient processors, and sooner or later, the gap will close, thus providing developers with enough processing power to implement better security features. However, I am not so sure this is a realistic scenario.

2.3. Insecure IoT:

First of all, IoT chips won't be big money-makers since they are tiny and usually based on outdated architectures. For example, the first-generation Intel Edison platform is based on Quark processors, which essentially use the same CPU instruction set and much of the design of the ancient Pentium P54C. However, the next-generation Edison microcomputer is based on a much faster processor, based on Atom Silvermont cores, which is in many Windows and Android tablets, today.

(Intel shipped ~46m Bay Trail SoCs in 2014.)

On the face of it, we could end up with relatively modern 64-bit x86 CPU cores in IoT devices, but they won't come cheap, they will still be substantially more complex than the smallest ARM cores, and therefore will need more battery power.

Cheap and disposable wearable, which appear to be the FTC's biggest concern, won't be powered by such chips, at least, not anytime soon. Consumers may end up with more powerful processors, such as Intel Atoms or ARMv8 chips, in some smart products, like smart refrigerators or washing machines with touchscreens, but they are impractical for disposable devices with no displays and with limited battery capacity.

Selling complete platforms, or reference designs for various IoT devices, could help chipmakers generate more revenue, while at the same time introduce more standardization and security. The last thing the industry needs is more unstandardized devices and more fragmentation. This may sound like a logical and sound approach, since developers would end up with fewer platforms and more resources would be allocated for security, however, security breaches would also affect a bigger number of devices.

Money Is Pouring In, Analysts Remain Bullish, What Could Possibly Go Wrong?

One of the most common ways of tackling any problem in the tech industry is to simply throw money at it. So, let's see where we stand right now in terms of funding rather than technology.

According to research firms IDC and Gartner, IoT will grow to such an extent that it will transform the data Centre industry by the end of the decade. Gartner expects the IoT market will have 26 billion installed units by 2020, creating huge opportunities for all parties, from data Centre and hardware makers, to developers and designers. IDC also expects the IoT industry to end up with "billions of devices and trillions of dollars" by the end of the decade.

Gartner's latest IoT market forecast published in May 2014 also includes a list of potential challenges, some of which I've already covered:

Security: Increased automation and digitization creates new security concerns.

Enterprise: Security issues could pose safety risks.

Consumer Privacy: Potential of privacy breaches.

Data: Lots of data will be generated, both for big data and personal data.

Storage Management: Industry needs to figure out what to do with the data in a cost-effective manner.

Server Technologies: More investment in servers will be necessary.

Data Centre Network: WAN links are optimized for human interface applications, IoT is expected to dramatically change patterns by transmitting data automatically.

All these points (and more) must be addressed sooner or later, often at a substantial cost. We are no longer talking about tiny IoT chips and cheap toys based on such chips, this is infrastructure. This is a lot of silicon in server CPUs, expensive DDR4 ECC RAM and even bigger SSDs, all housed in expensive servers, in even bigger data centers.

That's just the tip of the iceberg; industry must tackle bandwidth concerns, data management and privacy policies, and security. So how much money does that leave for security, which is on top of Gartner's list of IoT challenges?

A lot of money is already pouring into the industry, VCs are getting on board and the pace of investment appears to be picking up. There were also a number of acquisitions, often involving big players like Google, Qualcomm, Samsung, Gemalto, Intel and others. There is a list of IoT-related investments on Posts capes. The trouble with many of these investments, especially those coming from VCs, is that they tend to focus on "shiny" things, devices that can be marketed soon, with a potentially spectacular ROI. These investments don't do much for security or infrastructure, which would basically have to trail IoT demand.

Big players will have to do the heavy lifting, not VC-backed startups and toymakers. Agile and innovative startups will certainly play a big role by boosting adoption and creating demand, but they can't do everything.

So, let's think of it this way, even a small company can build a car, or tens of thousands of cars, but it can't build highways, roads, petrol stations and refineries. That same small company can build a safe vehicle using off-the-shelf technology to meet basic road safety standards, but it couldn't build a Segway-like vehicle that would meet the same safety standards, nor could anyone else. Automotive safety standards could never apply to such vehicles, we don't see people commuting to work on Segway, so we cannot expect the traditional tech security standard to apply to underpowered IoT devices, either.

Having commuters checking their email or playing Candy Crush while riding their Segway through rush hour traffic does not sound very safe, does it? So why should we expect IoT devices to be as safe as other connected devices, with vastly more powerful hardware and mature operating systems? It may be a strange analogy, but the bottom line is that IoT devices cannot be expected to conform to the same security standards as fully-fledged computers.

There Weren't That Many IoT Security Debacles…

True, we don't see a lot of headlines about spectacular IoT security breaches, but let me put it this way: how many security related headlines did you see about Android Wear? One? Two? None? It is estimated there are fewer than a million Android Wear devices in the wild, so they're simply not a prime target for hackers, or a subject for security researchers.

How many IoT devices do you own and use right now? How many does your business use? That's where the "Internet of No Things" joke comes from, most people don't have any. The numbers keep going up, but the average consumer is not buying many, so where is that growth coming from? IoT devices are out there and the numbers are booming, driven by enterprise rather than the consumer market.

Verizon and ABI Research estimate that there were **1.2 billion** different devices connected to the internet last year, but by 2020, they expect as many as 5.4 billion B2B IoT connections.

Smart wristbands, toasters and dog collars aren't a huge concern from a security perspective, but Verizon's latest IoT report focuses on something a bit more interesting: enterprise.

The number of Verizon's machine-to-machine (M2M) connections in the manufacturing sector increased by 204 percent from 2013 to 2014, followed by finance and insurance, media and entertainment, healthcare, retail and transportation. The Verizon report includes a breakdown of IoT trends in various industries, so it offers insight into the business side of things.

The overall tone of the report is upbeat, but it also lists a number of security concerns. Verizon describes security breaches in the energy industry as "unthinkable," describes IoT security as **"paramount"** in manufacturing, and let's not even bring up potential risks in healthcare and transportation.

2.4. How And When Will We Get A Secure Internet of Things?

I will not try to offer a definitive answer on how IoT security challenges can be resolved in This Topic or Section, or when. The industry is still searching for answers and there is a long way to go. Recent studies indicate that the majority of currently available IoT devices have security vulnerabilities. HP found that as many 70 percent of IoT devices are vulnerable to attack.

While growth offers a lot of opportunities, IoT is still not mature, or secure. Adding millions of new devices, hardware endpoints, billions of lines of code, along with more infrastructure to cope with the load, creates a vast set of challenges, unmatched by anything we have experienced over the past two decades.

I don't believe the industry can apply a lot of security lessons to IoT, at least not quickly enough, not over the next couple of years. In my mind, the Internet analogy is a fallacy, simply because the internet of the nineties did not have to deal with such vastly different types of hardware. Using encryption and wasting clock cycles on security is not a problem on big x86 CPUs or ARM SoCs, but it won't work the same way with tiny IoT devices with a fraction of the processing power and a much different power consumption envelope.

More elaborate processors, with a bigger die, need bigger packaging and have to dissipate more heat. They also need more power, which means bigger, heavier, more expensive batteries. To shave off weight and reduce bulk, manufacturers would have to resort to using exotic materials and production techniques. All of the above would entail more R&D spending, longer time-to-market and a bigger bill of materials. With substantially higher prices and a premium build, such devices could hardly be considered disposable.

2.5. The internet of things – IoT itself

So, what has to be done to make IoT secure? A lot. And everyone has a role to play, from tech giants to individual developers.

Let's take a look at a few basic points, such as what can be done, and what is being done, to improve IoT security now:

- Emphasize security from day one
- Lifecycle, future-proofing, updates
- Access control and device authentication
- Know your enemy
- Prepare for security breaches

A clear emphasis on security from day one is always a good thing, especially when dealing with immature technologies and underdeveloped markets. If you are planning to develop your own IoT infrastructure, or deploy an existing solution, do your research and stay as informed as possible. This may involve trade-offs, as you could be presented with a choice of boosting security at the cost of compromising the user experience, but it's worth it as long as you strike the right balance. This cannot be done on the fly, you have to plan ahead, and plan well.

In the rush to bring new products and services to market, many companies are likely to overlook long-term support. It happens all the time, even in the big leagues, so we always end up with millions of unpatched and insecure computers and mobile devices. They are simply too old for most companies to bother with, and it is bound to be even worse with disposable IoT devices. Major phone vendors don't update their software on 2-3-year-old phones, so imagine what will happen with $20 IoT devices that might be on your network for years. Planned obsolescence may be a part of it, but the truth is that updating old devices does not make much financial sense for the manufacturer since they have better things to do with their resources. Secure IoT devices

would either have to be secure by design and impervious from the start, or receive vital updates throughout their lifecycle, and I'm sure you will agree neither option sounds realistic, at least, not yet.

Implementing secure **access control** and device authentication sounds like an obvious thing to bring up, but we are not dealing with your average connected device here. Creating access controls, and authentication methods, that can be implemented on cheap and compact IoT devices without compromising the user experience, or adding unnecessary hardware, is harder than it seems. As I mentioned earlier, lack of processing power is another problem, as most advanced encryption techniques simply wouldn't work very well, if at all. In a previous post, I looked at one alternative, outsourcing encryption via the block chain technology; I am not referring to the Bit coin block chain, but similar crypto technologies that are already being studied by several industry leaders.

Si vis pacem, para bellum – "*if you want peace, prepare for war.*"

It is vital to study threats and potential attackers before tackling IoT security. The threat level is not the same for all devices and there are countless considerations to take into account; would someone rather hack your daughter's teddy bear, or something a bit more serious? It's necessary to reduce data risk, keep as much personal data as possible from IoT devices, properly secure necessary data transfers, and so on. However, to do all this, you first need to study the threat.

If all else fails, at least be prepared for potential security breaches. Sooner or later they will happen, to you or someone else (well, preferably a competitor). Always have an exit strategy, a way of securing as much data as possible and rendering compromised data useless without wrecking your IoT infrastructure. It is also necessary to educate customers, employees and everyone else involved in the process about the risks of such breaches. Instruct them in what to do in case of a breach, and what to do to avoid one.

3. Scope of IoT

3.1. Internet of Things (IoT): A Network of Everything

Today wherever we go, we hear a term called "IoT" which is creating more buzz in the list of emerging technologies in embraced by many big brands, consumers, and startups paving way for lots of innovation in technology.

On a lazy morning, our biggest problem is waking up from bed to on/off the fan. Luckily, we have got the solution for this kind of problems in the form of IOT. IOT may sound new to many, but we have already leveraged IOT in our daily life.

Many things around us are connected to each other through Internet of Things. It has been here for many years. Internet of Things History shows the complete history from the beginning to the present. It states the concept of Internet of Things started back in 1832 when an electromagnetic telegraph was created by Baron Schilling in Russia.

3.2. IoT-interest

Internet of Things can be defined as a network that allows things around us to interconnect and talk to each other. We live in a smart world. Things around us are smart, smart watch, smart T.V, smart mobile and more. The future of IOT is vast, We can experience it from the minute we wake up. Let me explain how IOT can bring magic into our lives in the near future.

Your Alarm may get synced with your calendar and wakes you up accordingly. Once you wake up it may quickly send a signal to your water heater and your water heater is ready with the hot water, that too would have adjusted the heat according to the temperature.

By the time you finish the bath, your coffee machine has already got the signal and busy in preparing your favorite coffee. Meanwhile, you have already alerted your car with your smartphone and by the time you lock your door it is waiting at your gate. Now you are all set to go. Wait, your refrigerator has sent you an alert that you are running out of milk in the refrigerator.

This sounds like a daydream, but most of this will come into existence in near future with Internet of Things. Here is the list of some already available IOT devices for making your home a smart home.

- Amazon Echo Voice controller
- Google Home Voice controller
- Coffee Smart Coffeemaker
- June Intelligent oven
- Ring Smart Video Doorbell
- Not only at home, IOT has potential to bring changes in the Business, Healthcare, Education and all other industries.

3.3. How IoT influences each industry:

3.3.1. Retail:

The interconnection of millions of devices in real-time will make the collection of valuable data and provide real-time insights for merchants. Key applications of IOT in retail industry also include supply chain and smart store applications.

3.3.2. Healthcare:

IOT can bring a big revolution in the process of treatment and diagnosis of diseases. Embedding the medical equipment with IOT devices will help to monitor the patients more effectively. Personal fitness and wellness trackers are already gaining popularity. Remote patient monitoring will also become more efficient with IOT.

3.3.3. Automobiles:

IOT connected smart cars are already in the market. Driverless cars and other automobiles are going to rule the future. Already many big companies are pouring money into this IOT connected smart cars and developed them. Soon we might see them on our roads. A study by BI expects that there will be around 94 million connected cars by 2021.

Internet of Things Wearable devices is the future. You can control your car with a smartwatch like automatically parking it and for closing or to open the doors. Your wearable device may also detect the dangers around you. We already have devices that track our activity and exercise habits. In near future, we may also be able to easily track our calorie intake and eating habits automatically. So, everything is going to be easy except that making everything easy is tough.

Let's discuss more on wearable devices in my next blog. Here are some examples of IOT wearable devices that are already available in the market.

- Sync Smart band Activity Tracker
- Fitbit Surge Smart Watch
- Hicon Wearable Smart Bracelet
- Beddit Sleep tracker
- Wearable devices

This clearly shows that the IOT is the next big thing. Many companies have already created path-breaking products using the Internet of Things. Not only for big companies IoT has provided many big opportunities for many start-ups too. Internet of Things – Top Opportunities for startups article provides you the information about the fast-growing Internet of Things startups.

3.4. Challenges we have to deal:

Despite huge opportunities and possibilities in IOT. There are some challenges involved like complexity, Privacy and Security issues, Data storage Issues and some more. But there are also ways with which we can overcome them. Block chain also plays a major part in the Internet of Things it enhances the security, makes transactions more seamless and creates efficiencies in the supply chain.

3.4.1. Security & Privacy concerns:

With IOT huge numbers of devices are connected to each other. Even, one poorly designed device can cause the security issues like data theft and possibilities of hacking the other devices will increase. This also raises the data privacy issues, since huge amount of personal data is collected by IOT devices. Sometimes the end user may not even be aware of what data the devices are collecting from them.

3.4.2. Stability:

Issues on how devices perform without internet or connectivity should also be considered when developing the devices. Making the device to perform in the similar way without the connectivity is also a challenge.

3.4.3. Compatibility:

Since IOT requires lot of devices to be interconnected the devices should be made compatible with each other. This needs the special hardware and software design.

3.4.4. Standards:

There are no proper standards followed or available for developing IOT devices. This may lead to the poorly designed and developed IOT devices which will in turn cause security and other major issues. At least in the future strong standards and best practices should be followed to develop IOT devices.

3.4.5. Intelligent Data Analysis:

A Huge amount of data should be analyzed to make the required actions. Even a small data flaws can cause the false assumptions. Handling unstructured data is also a big challenge.

The possibilities are endless! Internet of Things has already started transforming different industries and how they do their business. Industries like Manufacturing, Transport, Agriculture, Healthcare, Retail, Infrastructure, Logistics, Banks etc are going to get benefited hugely from Internet of Things. Experts are focusing on using these IOT smart systems to better their business.

The size of Internet of Things is definitely going to increase massively in future. A research expected that approximately there will be more than 24 billion IoT devices on Earth by 2020. This article is to make you understand the potential of IOT and how it’s going to change the future. There is much you have to learn about IOT. At our company, we constantly research the new possibilities in IOT and gained enough knowledge to build the next big product in IoT.

4. Attacks On IoT Based Devices

The Internet of Things (IoT) may be introducing a lot of benefits to modern life, but it also has one huge drawback: security threats.
In its 2018 IoT Predictions, Forrester Research noted, "Security vulnerabilities are a significant worry for firms deploying IoT solutions – in fact, it's the top concern of organizations looking at deploying IoT solutions. However, most firms don't consistently mitigate IoT-specific security threats and business pressures overwhelm technology security concerns."

The IoT security risks could be even more significant on the consumer side, where individuals are often unaware of the potential threats and what they should be doing to mitigate them. A 2017 survey on The State of IoT Security sponsored by security vendor Gemalto found that only 14 percent of the consumers surveyed considered themselves knowledgeable about IoT security. That number is especially concerning given that 54 percent of respondents owned an average of four IoT devices.
And those IoT security threats aren't merely theoretical. Hackers and cybercriminals have already found ways to compromise many IoT devices and networks, and experts say successful attacks are likely to increase. Forrester has predicted, "In 2018, we'll see more IoT-related attacks . . . except they'll grow bigger in scale and impact."
What type of IoT security threats will enterprises and consumers face in 2018? Based on historical precedent, here are ten of the most likely types of attacks.

Hackers have already proven that many of these IoT security attacks are possible.

4.1. Botnets and DDoS Attacks:

Many people in the general public first became aware of IoT security threats when they heard about the Mirai botnet in September 2016. By some estimates, Mirai infected approximately 2.5 million IoT devices, including printers, routers and Internet-connected cameras. The botnet creators used it to launch distributed denial of service (DDoS) attacks, including an attack against cybersecurity blog Krebson Security. Essentially, the attackers used all the devices infected with Mirai to attempt to connect to the targeted website at the same time, in the hopes of overwhelming the servers and preventing anyone from reaching the site.
Since Mirai first made news, attackers have launched other IoT botnet attacks, including Reaper and Hajime. Experts say more such attacks are likely in the future.

4.2. Remote Recording:

The possibility that attackers could hack into IoT devices and record the owners without their knowledge came to light not as a result of the work of hackers, but as a result of the work of the Central Intelligence Agency (CIA). Documents divulged by WikiLeaks implied that the spy agency knew about dozens of zero-day exploits for IoT devices but did not disclose the bugs because they hoped to use the vulnerabilities to secretly record conversations that would reveal the activities of US adversaries. The documents pointed to vulnerabilities in smart TVs, as well as Android and iOS smartphones. The obvious implication is that criminals could also be exploiting these vulnerabilities for nefarious purposes.

4.3. Spam:

In January 2014, one of the first-ever known attacks involving IoT devices used more than 100,000 Internet-connect devices, including TVs, routers, and at least one smart refrigerator to send 300,000 spam emails per day. The attackers sent no more than 10 messages from any one device, making it very hard to block or pinpoint the origin of the attack.

This early attack was far from the last. IoT spam attacks were continuing last fall with Linux.ProxyM IoT botnet.

4.4. APTs:

In recent years, advanced persistent threats (APTs) have become a major concern for security professionals. APTs are highly capable attackers, such as nation-states or corporations, that launch sophisticated cyberattacks that are difficult to prevent or mitigate. For example, the Stuxnet worm that destroyed Iranian nuclear centrifuges and the 2014 Sony Pictures hack have been attributed to nation-states.

As more critical infrastructure gets connected to the Internet, many experts warn that APTs could launch an IoT attack targeting the power grid, industrial control systems or other Internet-connected systems. Some have even warned that terrorists could launch an IoT attack that could cripple world economies.

4.5. Ransomware:

Ransomware has become all too prevalent on home PCs and corporate networks. Now experts say it is just a matter of time before ransomware attackers start locking up smart devices. Security researchers have already demonstrated the ability to install ransomware on smart thermostats. They could, for example, turn up the heat to 95 degrees and refuse to turn it back to normal until the owner agreed to pay a ransom in Bitcoin. They might also be able to launch similar attacks against connected garage

doors, vehicles or even appliances. How much would you pay to unlock your smart coffee pot first thing in the morning?

4.6. Data Theft:

Obtaining sensitive data, such as customer names, credit card numbers, social security numbers and other personally identifiable information, continues to be one of the primary goals of cyberattacks. And according to the Ponemon Institute, the average data breach costs companies $3.62 million, or about $141 per record stolen. IoT devices represent a whole new attack vector for criminals looking for ways to invade corporate or home networks. For example, if an improperly secured IoT device or sensor is connected to enterprise networks that could give attackers a new way to enter the network and potentially find the valuable data they are looking for.

4.7. Home Intrusions:

As smart locks and smart garage door openers become more commonplace, it also becomes more likely that cybercriminals could become real-world thieves. Home systems that are not properly secured could be vulnerable to criminals with sophisticated tools and software. Disturbingly, security researchers have demonstrated that it is fairly easy to break into smart locks from several different manufacturers, and smart garage doors don't seem to be much safer.

4.8. Communicating with Kids:

One of the most disturbing stories of IoT security gone wrong involved hacking a baby monitor. One couple discovered that a stranger had not only been using their baby monitor to spy on their three-year-old son, that stranger had also been speaking with their child over the device. The mother heard an unknown voice say, "Wake up little boy, daddy's looking for you," and the child said that he was scared because someone was talking to him over the device at night.

As more children's gear and toys become connected to the Internet, it seems likely that these frightening scenarios could become more commonplace.

4.9. Remote Vehicle Control:

As vehicles become smarter and gain connections to the Internet, they also become vulnerable to attack. Hackers have shown that they could take control of a Jeep, setting the air conditioning to maximum, changing the radio station, starting the wipers and eventually slowing the vehicle to a stop. The news led to a recall of 1.4 million vehicles, but the white-hat researchers behind the original exploit said that they found

additional vulnerabilities that weren't addressed by the patch Chrysler applied to the recalled vehicles. Although experts say that the auto industry is doing a better job of securing vehicles, it's almost certain that attackers will find new vulnerabilities in connected cars.

4.10. Personal Attacks:

Sometimes the IoT encompasses more than just things — it can also include people who have connected medical devices implanted in their bodies. An episode of the television series Homeland featured an assassination attempt that targeted an implanted medical device, and former Vice President Dick Cheney was so worried about such a scenario that he had the wireless capabilities on his implanted defibrillator disconnected. This type of attack hasn't happened in real life yet, but it remains a possibility as more medical devices become part of the IoT.

5. Vulnerabilities in IOT

Kevin Ashton instituted the expression the "Internet of Things" (otherwise called "IoT") in 1999. This term spoke to the idea of a monstrous framework where regular machines would speak with one another through ubiquitous sensors.
From that point forward, purchasers have associated keen gadgets among each other at an exponential rate. By 2020, there are gauges that there will be seven shrewd gadgets for each individual. That implies a future where numerous ordinary things will interconnect with one another, with the essential objective of enhancing the general personal satisfaction.

As we enter the new age of the Web of Things, we should reconsider the whole thought of security. The IoT is making a huge difference every day. In this way, a substantial number of people and associations are obtaining and executing gadgets to enhance our lives and dimensions of efficiency.

For example, through the IoT we can consider the perfect world; we remotely see that grandpa is moving about the house, his heart is alive and well, and he has milk and nourishment in the icebox.

In any case, the real concern is that webcam in the carport may have possibly been a piece of a dispersed Forswearing of Administration assault. Without a doubt, our reality is developing. The IoT carries with it a phenomenal future; at the same time, we ought to recollect the various security and insurance contemplations which should be thought about.

“Nothing is useless and nothing is perfect”

5.1. Ten Basic Internet of Things Security Vulnerabilities

There are many assault vectors related with IoT gadgets. Since huge numbers of them are Digital based, it is along these lines very difficult to verify and deal with a general IoT foundation. In this article, we will consider ten IoT vulnerabilities that exist today.

5.1.1. A Precarious web interface

Various contraptions and gadgets have a worked in web server that has a web application for overseeing them. Like any web server/application, there may be defects in the source code that reason the interface to be defenseless against a Digital based assault.

5.1.2. Inappropriate Utilization of Confirmation or Approval Systems

Regularly in IoT gadgets, there are blemishes in the execution of the approval/validation instruments. It deteriorates when the gave security includes in the IoT gadgets are not used by the customers to their fullest degree conceivable.

5.1.3. Uncertain system administrations

IoT gadgets have devices for diagnostics and testing alongside administrations like investigating. These "upkeep" administrations have perhaps been daintily tried, which in this way make them prone to have exploitable source code behind them. At the end of the day, more highlights mean greater security defects in the IoT gadgets and their individual framework.

5.1.4. A nonattendance of transport layer encryption

On the off chance that your IoT gadget sends private information over a shaky convention, it is in cleartext, and therefore, anyone can peruse it. This just underscores the significance have secure interchanges conventions with these gadgets.

5.1.5. Protection issues

In the event that the information on the IoT gadget isn't encoded, and different people approach it, this makes your information helpless against undercover capturing and robbery.

5.1.6. Untrustworthy cloud interface

Various IoT gadgets can associate with the Cloud. Consequently, having a Cloud-based administration interface speaks to one more potential security helplessness.

Hence, an on-gadget the executive interface is considerably more troublesome for a remote aggressor to get to on the grounds that it is behind the home switch or firewall.

5.1.7. Untrustworthy versatile interface

Cell phones are utilized for nearly everything nowadays. Subsequently, IoT gadgets have a portable interface too. Be that as it may, having another administration interface implies one more road for the Digital aggressor to get into the IoT framework.

5.1.8. Lacking security highlights

Commonly, IoT gadget might not have the security highlights incorporated with them as it is professed to be. Thus, there are no encryption choices or highlights to recognize and screen any potential Digital assaults.

5.1.9. Unprotected fixes and redesigns

In such manner, these inquiries should be expressly tended to: Will the IoT gadgets be fixed to address any security vulnerabilities? How would I know whether I require a fix? Will presenting a particular fix further effect the gadget being referred to? How might I know whether the fix is authentic and not imbued by malware or spyware?

5.1.10. Deleting individual information

This one isn't on the OWASP list yet, however it to seem, by all accounts, to be a critical issue in IoT gadgets. For instance, on the off chance that I need to exchange my IoT gadget, what steps would I be able to take to ensure that my own data and information that live in it is completely cleaned away before another client approaches it.

SO

“It is fundamental to consider these security vulnerabilities when you are acquiring an IoT gadget. In the event that you are an IoT designer, at that point give genuine thought regarding how you can plan progressively powerful and secure source code for those gadgets that you communicate with every day. In any case, the distinct the truth is that customers will continue acquiring IoT gadgets dependent on value, ease of use, and additional highlights. Therefore, security contemplations could take the famous "back burner" for a long while to come.”

CHAPTER 2

IOT BASED SMART CAMPUS SYSTEM

This section describes the development of smart campus using Internet of Things (IoT) technology. Through smart campus, it is possible that a campus is connected via online by the outside entity, so that the teaching approach based on technology can be conducted in real time. Many of researches have conducted already in smart education, smart parking and smart room. The result of this research is the design of smart campus system that includes smart education development, smart library, smart cafeteria, smart lighting, smart ID card system, smart parking and smart room like system with the sake of Qarshi University as the case study.

1. IoT Based Smart Campus System

Smart Campus is an entity of any kind that uses technology and infrastructure to support and improve its processes, so people can use them better. Our thesis reviews the literature to contextualize the Internet of Things and its vital importance for the Smart Campus, as well as its relationship with the concepts of cybersecurity and wireless sensor network. We describe the various interrelationships, tendencies, and future development of a Smart Campus, as well as the differences and similarities with the emerging concept of Smart University. This review revealed that the Internet of Things is involved in all fields and can influence and improve the university's processes to contribute to decision-making, technological development, and academic learning. To conclude, Smart University focuses on improving the infrastructure of universities through technology, with the main purpose of enhancing the quality of the education provided by institutions.

The research by proposes the solution of RFID design structure integrated by using modern technology from cloud computation, supported by good quality technology and economy. This technology supports things to increase the campus security, asset track record, valuable things, student record, security of paper exam and original certificate. The future class appears to increase pedagogy, in which the students can participate more in the learning process through interaction and collaboration. To make environment that supports future classroom, it needs integrated mechanism of any related things being needed. In this research, the design of structure of Internet of Things was built, in which application domain is being made as system unity related to the internet.

To fasten and develop Intelligent Campus Internet of Things (ICIoT) more efficiently, the approach based on the runtime model to manage campus was applied as the result of the use of Wireless Sensor Network (WSN). It is hoped that all of management job description can be conducted by executing program that the model has been made appropriate. In the learning part, experiment was conducted and compared by using the traditional method. This method can increase the management effectiveness of campus facilities, save energy as much as 16.7%. In the study conducted by, the concept of internet-based parking guide by using QR code was introduced. This system runs on cellular platform by visualizing the available parking lot for the customers so that they can make order.

1.1. Smart Campus Infrastructure

Smart Campus is one of the innovations that will be developed in UPY. Adopted from the existing smart city system, one of aspects that being considered in developing smart campus is the infrastructure. The infrastructure is the main key of the campus smart program. If the infrastructure has been well developed, the information related to the campus can be accessed from mobile phone or others gadgets. Some parameters of the smart campus that being adapted from smart city are:

a. Smart education consists of: eLearning, Personalized Learning, Virtual Classroom.
b. Smart parking, a parking system that provides the information about available parking lot and the information that the parking lot has been full.
c. Smart room, system that provides information related to the classroom that is being used or vacant.

1.2. Architecture of Smart Campus

1.2.1. Literatures Survey

The main contribution of this paper is an original proposal of an application, platform and data independent model for building the integrated information system that will support educational and other processes at University level. Increased number of students causes a necessity for more precise and efficient approaches of students' authentication and monitoring. based on the affordability and efficiency, the proposed method for students' identification is by using smart cards. The chip on a smart card can store multiple identification factors of a specific user: institution/faculty ID, user ID, password, grades etc. CMS is also used for providing environment, general information about the institution (university/faculties), public news and announcements (on the university/faculty level) etc. However, every user needs to provide a valid authentication data in order to access eLearning. We need to protect content, services and personal data from outside intruders and also these systems carry a risk of privacy violation from inside staff (administers and educational staff).

In this paper we divide the whole system into three layers. The IoT is import part of new information and it connects everything through RFID, sensors QR code and real-time positioning technology and realizes the intelligent identification of location and management for goods, IoT of education fully perceives the workers, resources and equipment's of school in a perceptual layer. Then the network layer is responsible for reliable transmission of information from perceptual layer, then IoT realizes the intelligent analysis, early warnings and intelligent scheduling in application layer. Here the isolated systems such as educational management system, finance management system and office management system are integrated by IoT technology. This paper concentrates on office management more than on education or research.

Here we focus on the campuses that are spread over fairly large area and it is difficult to control for management to track everything happened. This paper tells us about the need of adopting IoT technology in campus using secured smart system for campus academics. In this system sensors are enabled and network devices work

continuously and collaboratively to give humans more comfort. The smart classroom collects information, stores it as digitalized data in a memory of e-campus platform. Here platform is created for fascinating learning by means of smart classroom surroundings and security for e-campus. The paper mainly considers the security aspects leaving behind many other important features.

The major issue considered here is how different services can be integrated into our smart campus which is heterogeneity of systems and technologies that they use. The service provided here aim at increasing productivity in campus, saving a lot of time and making it easy for all people present in campus. Here we propose a system where smart cards are implemented for access control and payments and data generated is analyzed for human behavior. Smart garbage system is also established using simpler technologies. The proposed system consists of many aspects like smart micro grids, smart lighting, security and safety but all this are not implemented due cost constraint and other problems.

Here mainly we describe IoT enabled classrooms in which data collection can be made possible by using devices for e-learning application. The data is fetched in real time activity. The smart classroom data are stored in digital data eLearning platform. Student outside the classroom can access subject data notes using information sharing display board and it can be controlled by administrators for adding updates. Using IoT learning application provides simple and cost-effective notes to any corner of the world. Now we recommended the efficient way of designing smart learning through IoT. It thus probable's teamwork to rapidly increasing timely delivery of subject notes, easy way of learning and data reaches speedily. This paper focuses only on enhancing effective way of classroom teaching technique not towards smart campus.

Our objectives are to develop an application that provides a smart and easy way for execution of several academic operations to provide students with information regarding complaints any placement activities or any other event happening in the college. It addresses the problem of students not obtaining correct notifications at correct time, here the gap between college students and administrators are tried to be filled using android application. The main idea here is to design and develop an android based college management system. It reduces the physical human effort. The proposed system focuses only on education system neglecting other aspects that needs to be addressed to create a smart campus.

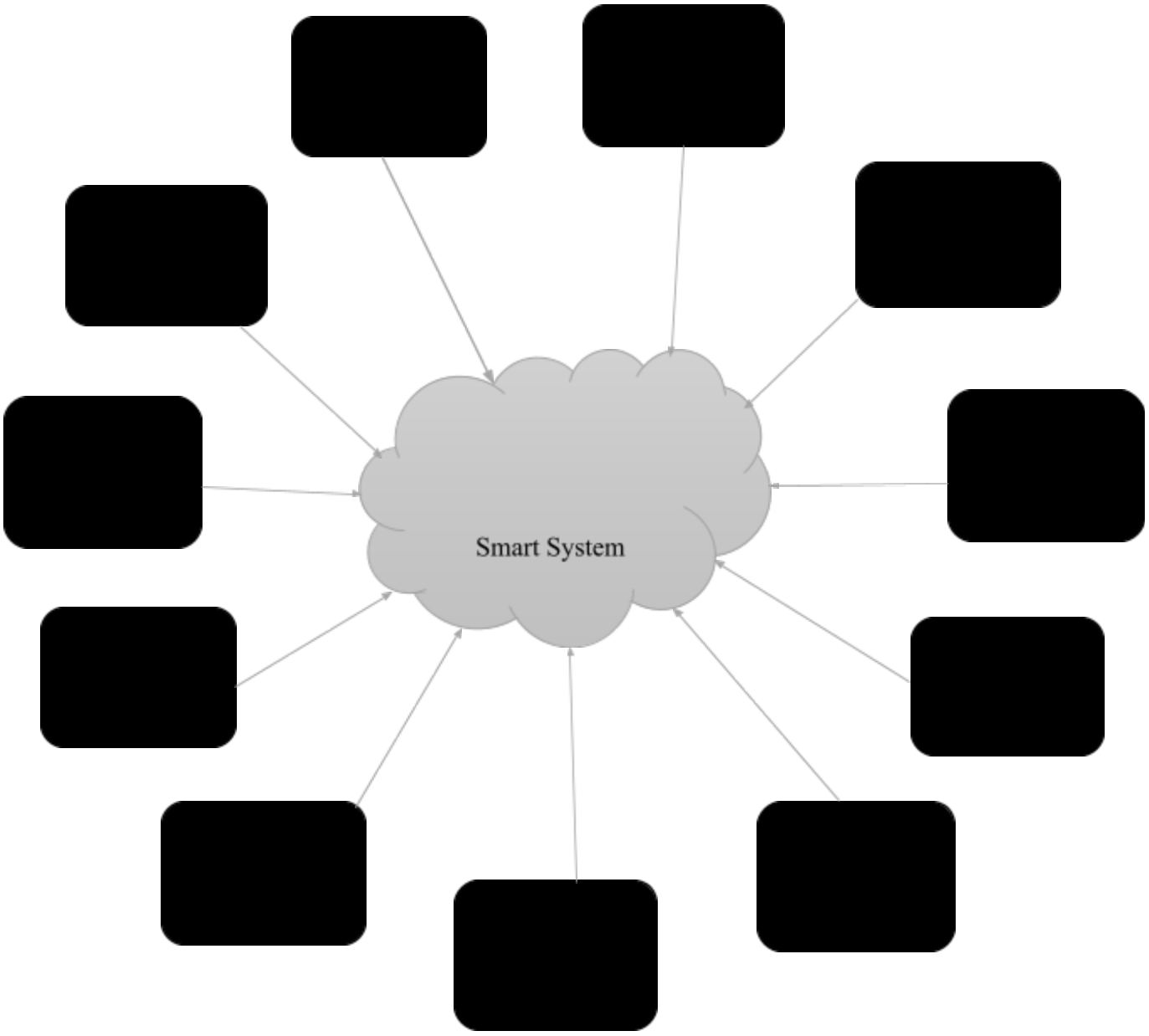
This section presents the up-to-date outcomes of research project that is aimed on analysis of students with disabilities & how they might benefit from smart software & hardware systems & smart technology. Smart system integrates voice recognition, computer-vision & other technology. The objectives for this particular research include, but are not limited to Identification of smartness levels in a smart education system. Identification of characteristics of students with various type of disability. Identification of software and hardware systems and technology to aid students with disabilities in highly technological SMCS. It indicates the difference between digital campus & smart campus and it connects everything through RFID, sensor, QR code & RT positioning technology & realize the intelligent identification, location & for goods. The system does not describe many of the aspects that make a complete smart campus.

Several classrooms, staff offices and laboratories etc. have been equipped with super sensors featuring temperature and light monitors. The factors like temperature and

light can have significant effects on the productivity of office workers as well as students. Also, this helps in reducing power consumption by monitoring the environment condition and controlling the appliances. An IR sensor in each room allows building users to find a meeting room that is currently free. Generally, public meeting rooms should be booked via a shared calendar system. These kinds of Systems are required in the university campus as the area is very large and number of rooms are also large. And human can make mistakes and forget to switch off the appliances when in no use and in this case, these systems are useful in order to increase the power efficiency. The system can be viewed as a future of artificial intelligence. This is a powerful and dependable system. It fulfills the goal of energy saving and helps in achieving the efficient use of energy resources. The section mainly focuses on the conservation of energy in the campus and not the actual smart campus system.

1.3. Proposed Methodology

This project uses multiple sensors, some sensors like soil moisture sensor, ultrasonic sensor and cameras are connected to a main processor which is Raspberry Pi and remaining application sensors like IR sensors mic and others are connected to the Arduino processor, the sensors will send a message to main processor for its proper working at the power up stage and processor will acknowledge its present. Each sensor sends the information to main module which either saves the information into a database or it simply proceeds with the required action, information to be saved is sent to Cloud/Local server and appropriate action or function that is carried out is also saved depending on the type of the data. For the functions like Smart canteen and Smart library which can be accessed by students will be operated using smart phone application. The device that connects the android application and cloud will Raspberry Pi's function known as FLASK. Each time students request will be processed and updated in database.


Smart System

1.3.1. Smart Building

The Internet of Things (IoT) is advancing a new breed of smart buildings that are better aligned with the priorities of property owners and managers. IoT enables operational systems that deliver more accurate and useful information for improving operations and providing the best experiences for tenants. Accelerating this transformation, we are designing an IoT building blocks that simplify how building systems talk to the cloud and exhaustively analyze building data.

Building managers need to provide their customers with competitive management and automation services in order to provide best-in-class climate control, energy savings, and maintenance. Converting existing buildings to control systems can be extremely challenging and costly because of the variety of legacy equipment and customer expectations.

What is needed is an open building automation platform capable of integrating the numerous building systems that today lack the connectivity and interoperability to work in unison. This design is making this a reality through easy-to-deploy hardware and software building blocks. These solutions improve building visibility and manageability, giving facilities organizations the tools, they need to reduce operational costs and provide the best experiences for tenants.

1.3.2. Smart Parking

Smart parking consists of IR sensors, cameras etc. When a vehicle arrives at the gate the IR sensors senses the vehicle, the camera captures image and arrival time of vehicle, then the programing is done in such a way a that LCD display shows using LED which directs the vehicle to its allotted lot. In the parking area, we again use IR sensors, when the sensor gives logical zero as an output then that particular which is used to set the threshold value.

Real-time views of parking availability and traffic conditions help students arrive to class on time. Connect campus buses, bike shares and vehicles to provide real-time location and arrival information to students' smartphones. Reduce traffic congestion by making it easy for people to find, reserve and wirelessly pay for parking.

Smart parking refers to the use of sensing devices to determine occupancy at the space level or at the lot/structure level. Sensing devices can refer to cameras, counting equipment like gates at the entrance of a lot, or sensors that are embedded into the pavement of individual parking spaces, to name a few.

When a vehicle arrives at the gate the IR sensors senses the vehicle, the camera captures image and arrival time of vehicle, then the programing is done in such a way a that LCD display shows which parking lot is empty there are also arrow marks shown to direct the vehicle to its respective parking lot.

1.3.3. Smart ID Card

Open up closed-loop student ID payment systems to integrate with credit and debit cards, and mobile payments. Integrate Smart ID cards with student information

systems to automate and digitize attendance, voting, bursary distributions and even access to residence halls and school facilities.

1.3.4. Smart Wayfinding

We can use real-time location services to make it easy to navigate and take advantage of campus life. Provide digital kiosks with interactive maps of campus facilities and services. Offer smartphone apps that trace a student or visitor's path to their destination. Deliver messages and promotions— "20 percent off today in the campus bookstore"—triggered by a student's proximity to the student union.

1.3.5. Smart Examination System

The exam statistics are crucial for instructors to better assess the exam metrics, e.g., exam duration. However, one statistic, time spent on each question by each student, is not available or difficult to obtain by human observation. In this work, we propose a smart exam system that obtains such statistics and sends data to a web application for further processing by using an internet-of-things based technology. The information obtained by the proposed system can even be used to diagnose students with learning difficulties or disabilities, e.g., dyslexia. We can make an experimental setup and conducted experiments with volunteers. The experiments conducted during the study show that the proposed system effectively collect information and process data for the use of instructor.

1.3.6. Smart Time Table

Scheduling of classes, availability of teachers in their office time, availability of the free lecture rooms and the scheduling of online devices connected to use in specific time period. All this can be done in smart timetable which should be embedded in the app so anyone either he is student or fecality can get to know about timings on time to time.

1.3.7. Smart Attendance

2.3.1 Barcode

In this technology, students carry barcodes printed on their student cards, which at the time

of presenting barcodes read using barcode reader, this presence is fast enough and does not
cost a fortune, but this system has a drawback where barcodes can be easily duplicated by
printed using a regular printer.

2
MATEC Web of Conferences 164, 01020 (2018) https://doi.org/10.1051/matecconf/201816401020
ICESTI 2017

2 Materials and methodology

2.1 Radio-frequency identification (RFID)

RFID has grown rapidly in recent decades [1] along with the demand from modern industry where data accuracy is required and efficiency improvements of a system, the supporting components of this electronic equipment are RFID reader and RFID Tags, where many kinds of tags have been manufactured according to industry needs. This technology has

been applied to various sectors such as industry [2], airports [3], attendance monitoring systems [4–8], and with the use of IoT will be able to make this system more optimal [9–11].

Fig. 1. Components of an RFID system [1].

Fig. 2. Type of an RFID tags.

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2.3.2 Magnetic Stripe

The use of magnetic strips can eliminate the deficiencies of barcodes, because in this
technology magnetic card is required as an ID for each student, in the duplication of the

magnetic card cannot be done easily because it requires a magnetic card reader and magnetic card writer, which because of the price quite expensive so it cannot be easily owned by students.

2.3.3 Biometric

The use of presences using biometric is an excellent solution, because as the ID of each student can be a finger print, retina, or face recognition. However, it has to consider the price of this equipment is very expensive.

Radio Frequency Identification (RFID) attendance monitoring system using RFID is highly recommended, because the required equipment is relatively cheap and the attendance process is very fast, so to make a monitoring system presence with this equipment can save time and budged.

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Using any of these technologies connected with cloud can be shift into smart attendance.

1.3.8. Smart Lab

IOT terminals, such as smart home devices and especially wireless accessible devices, are more popular than ever. These devices are called “smart” because they are more clever than normal single-chip devices. They always have an ARM-based CPU inside. They can easily be connected to a network, and they are always programmable and accessible. For this reason, a smart lab network laboratory is now a necessity for universities offering IT education. With the rapid development of IOT technology, PC programming courses alone are not enough. A smart lab course provides hardware and software tools that will help students construct a real IOT environment and develop in it. Without such a system, students may need several years to learn IOT concepts and master programming methods on different platforms.

The smart lab offers many benefits to a university. It allows everything to be under control. A smart lab server running in IDC (Internet Data Center) provides data access and remote operating services to any manager on the campus. It is a quick and convenient way to check or respond to any issues with devices on campus. The integrated desktop utilities allow managers to operate any devices remotely and immediately.

In a smart network laboratory managed by a smart lab system, students have the opportunity to learn and practice most of the technology used in an IOT. Network structure and program practices allow students to gain the basic and important knowledge of the Internet. Students will learn some key IOT concepts and technology, such as IOT node theory, wireless communication, sensors and triggers, and how to connect an IOT node to an IP network.

1.3.9. Smart Library

The library is one of the important parts in any educational organization. Although, library has a system, but the library needs to implement a new management system in order to replace the existing system by introducing the new system. There are many reasons why the library staffs have to implement another system, which are: loss a lot of information on the library books. The loss of data about the books Difficulty in tracking down the details of the library transactions due to a slow system Difficulty in updating the information on regular basis. Smart library by using the concepts of Radio Frequency Identification (RFID) and Mobile (IoT) the new system will manage and control all the information of the library and solve the above mention problems and as well as provide several benefits for the staff & students. In application it includes several forms which will be used by the library staff and students.

Smart Library system is combined with android application that is created and is installed in all the student phone. This application will inform all the students about the books available in the library, question papers and other books, magazine information. This android application will also inform as about the status of books that has be borrowed. In the library when the student enters it is monitored by cameras and the student when enter the aisle in which the required book is present is supposed to swipe the identity card only then that aisle will be open allowing the access to obtain the books. The books each contain NFC tags (Near Field Communication) which will have its reader, when a particular book is removed from the aisle it is accessed from the database which NFC tag and its number belonging to when books, is corresponding data is retrieved. The student is given 30 seconds time to return the book to its place. Before its

get added to its account, at the end of retrieving the book the student must swipe the book in front of NFC card reader so that the book can be added to that student account and be removed from the available book list. If the book which is removed is not exposed to NFC card reader by the student, then when walking out and alarm is set to alert the authorities of an authorized book retrieval. The data base is created in MySQL software which will allows us to store that database in cloud for easy access. The store information can be retrieved as and when required. The MySQL database for smart library consists of all the information of the book present in library, student information and staff information. The books that are exposed to NFC card reader is removed from available list and student information linked and informed to authorities. When the number of days allotted to keep that book is up, a notification warning is sent to the student to return the book and if book is not returned on time then the amount corresponding is reduced

1.3.10. Smart Cafeteria

Smart Canteen will contain the list of food available that particular time. In application this can be accessed by both student and other faculty members, through application they would also place their order for the food they want. The amount of the order food will be deducted after 10 minutes of placing order which is the time deduction given to the customer cancel their order. The list of food items is updated on the database daily which is also accessed from cloud. Cloud used here is AWS cloud (Amazon Web Services) at the preparator end their will be screen showing in the list of orders for food placed. The preparators will prepare the food ordered in the given delay of 10 minutes once the food is prepared there will be a buzzer indication the order number i.e. ready for service. The list of customers is also storing with the amount of food ordered to keep the accounts list of food items are updated every day as the change in menu and same is updated in database.

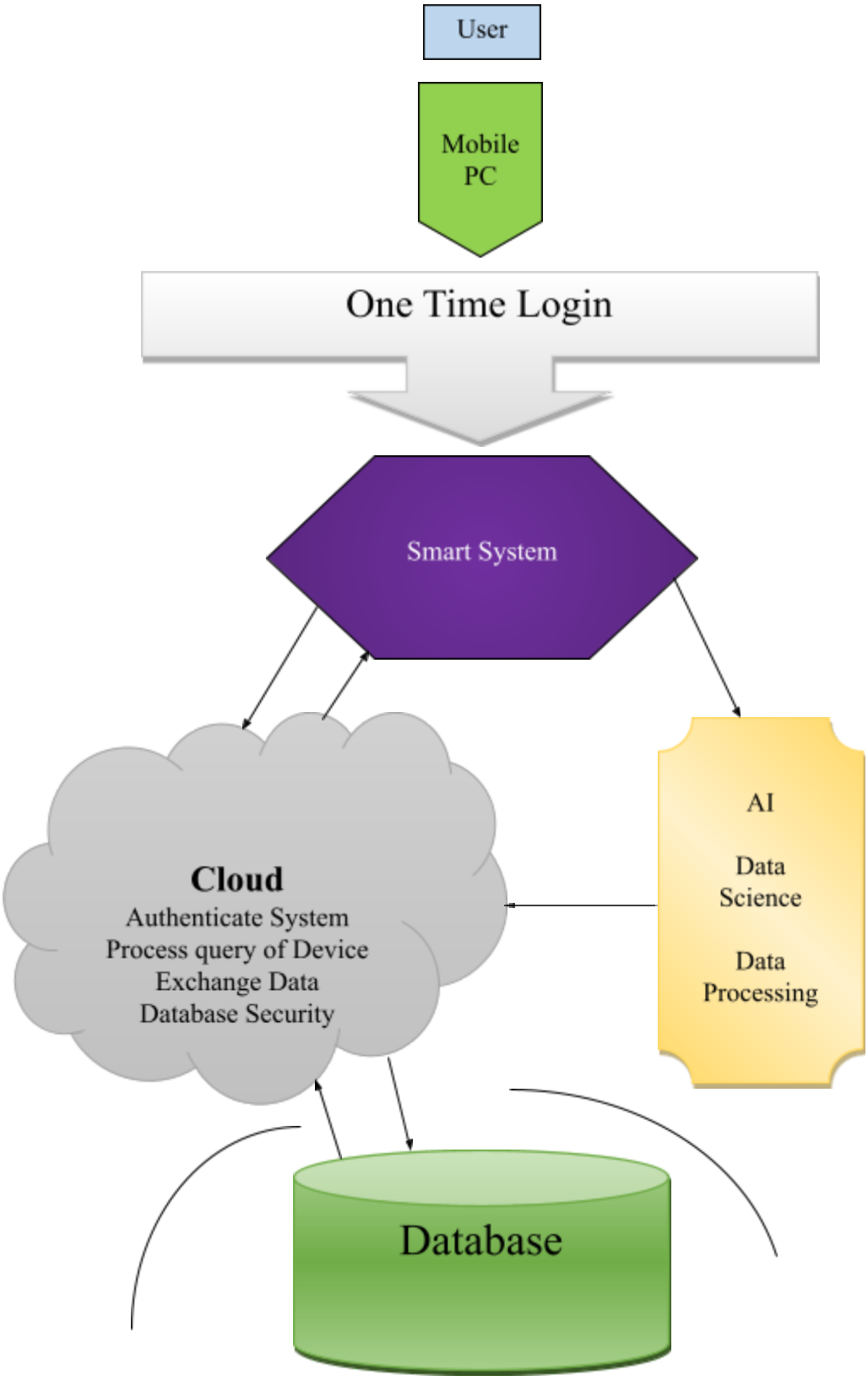
1.3.11. Smart Lighting System

Here the system is realized using the IR sensor (Infrared). IR sensors have range of 2-30cm these IR sensors are connected on street lights depending on the presence or the movement. The light brightness if there is a no presence it is coded in such a way that the light is dimed this is done so that power can be saved.

It comes in Smart Automation consists of controlling, when the light and the fans will turn ON or OFF, her we are using PIR (passive infrared sensors) which is used for motion detection, PIR sensors can detect motion its range of 120 degrees and up to 7 meters. When motion is detected the sensors gives logical high as its output and the fans and lights are turned on, once there is no motion logical zero output is obtained, fans and lights are turned OFF.

This project aims for designing and executing the advanced development in embedded systems for energy saving of street lights. Currently we have a manual system where the street lights will be switched ON in the evening before the sunsets and they are switched OFF in the next day morning after there is sufficient light on the outside. But the actual timing for these lights to be switched ON is when there is absolute darkness. With this, the power will be wasted up to some extent. This gives solution for electrical

power wastage. Also, the manual operation of the lighting system is completely eliminated. The proposed system provides a solution for energy saving. This is achieved by sensing and approaching a vehicle using an IR transmitter and IR Receiver couple. Upon sensing the movement, the sensor transmits the data to the microcontroller which furthermore the Light to switch ON. Similarly, as soon as the vehicle or an obstacle goes away the Light gets switched OFF as the sensor sense any object at the same time the status (ON/OFF) of the street light can be accessed from anywhere and anytime through internet. This will be implemented with smart embedded system which controls the street lights based on detection of vehicles or any other obstacles on the street. Whenever the obstacle is detected on the street within the specified time the light will get automatically ON/OFF according to the obstacle detection and the same information can be accessed through internet. The real time information of the street light (ON/OFF Status) can be accessed from anytime, anywhere through internet.


User
Mobile
PC
One Time Login
Smart System
Cloud
Authenticate System
Process query of Device
Exchange Data
Database Security
AI
Data
Science
Data
Processing
Database

CHAPTER 3

Threats and Attacks

1. IoT Security, Privacy, Threats, Attacks and Challenges

The era of IoT has changed our living styles. Although the IoT provides huge benefits, it is prone to various security threats in our daily life. The majority of the security threats are related to leakage of information and loss of services. In IoT, the security threats straightforwardly are affecting the physical security risk. The IoT consists of different devices and platform with different credentials, where every system needs the security requirement depending upon its characteristics. The privacy of a user is also most important part because a lot of personal information is being shared among various types. Hence a secure mechanism is needed to protect the personal information. Moreover, for IoT services, there are multiple types of devices that perform communication using different networks. It means there are a lot of security issues on user privacy and network layer. User privacy can also be uncovered from different routes. Some security threats in the IoT are as follows:

1) **E2E Data life cycle protection:**
   To ensure the security of data in IoT environment, end-to-end data protection is provided in a complete network. Data is collected from different devices connected to each other and instantly shared with other devices. Thus, it requires a framework to protect the data, confidentiality of data and to manage information privacy in full data life cycle.
2) **Secure thing planning:**
   The interconnection and communication among the devices in the IoT vary according to the situation. Therefore, the devices must be capable of maintaining security level. For example, when local devices and sensors used in the campus network to communicate with each other safely, their communication with external devices should also work on same security policy.
3) **Visible/usable security and privacy:**
   Most of the security and privacy concerns are invoke by misconfiguration of users. It is very difficult and unrealistic for users to execute such privacy policies and complex security mechanism. It is needed to select security and privacy policies that may apply automatically.

1.1. Security system structure

Security system structure of smart campus while bringing convenience to smart campus, campus cloud also faces a series of security problems. Besides the long-existing information security problem, there are also new security problems caused by cloud computing, such as scattered storage of data under the environment of cloud computing and openness of cloud service.

All this brings new challenges to the existing security system. Its security concerns whether cloud computing can be applied in smart campus. According to the practical situation in combination with characteristics and technologies of cloud computing, corresponding to cloud computing architecture, security system structure of smart campus contains six layers, i.e. physical security, network security, data security, service security, application security and management security.

1.1.1. Physical security

Physical security is the basis and premise of guaranteeing safe operation of smart campus. This section shows how to improve physical security mainly in three aspects environment, equipment and medium.

1) Environmental security:

Site facilities such as machine room, computer equipment and monitoring and ambient environment and fire safety should meet relevant national standards; machine room should keep away from various high risk (earthquake, magnetic field, lightning and fire disaster) regions; when the system is damaged, it should have corresponding forecasting, alarm and automatic damage elimination mechanism; the system should not only have complete fault tolerance measures and single-point fault correcting measure; to prevent electromagnetic leakage, 1243 internal equipment of the system should use technologies such as shielding and ant-interference.

2) Equipment security:

To guarantee safe operation of system equipment, power and electrostatic protection technologies, anti-virus, anti-electromagnetic, anti-short circuit/open circuit technologies etc. should be used. Equipment operators should receive corresponding education, training and management. There should be a reasonable man-machine communication mechanism so as to avoid abnormal operation of equipment to a great extent and improve the efficiency and lifetime of equipment.

3) Medium security:

When the use of transmission medium of smart campus network is considered, optical fiber should be used. It is sealed in plastics and placed underground, which can prevent open circuit/short circuit and parallel wiretap.

1.1.2. Network Security

Security system structure of smart campus Network security All applications of smart campus provide services based on network. Therefore, the reliability of network security of campus cloud becomes the key to normal and continuous service provision. Network security is guaranteed mainly in such aspects as network topology security, partition of security domain and boundary protection, data flow load balancing, access control of network resources, means of intrusion detection and network facility anti-virus.

Network topology security uses new network topological structure which enables higher connectivity of nodes and fault tolerance ability and easier load balancing. In campus cloud, physical security boundary gradually disappears and is replaced by logic security boundary. Virtual private network and data encryption technologies should be adopted to ensure transmission security of user data. To prevent excessive concentration of data flow which causes network congestion, campus cloud system should be able to analyze the load level of server, correctly guide user requests according to load level, automatically choose those stable and reliable networks and realize load balancing between servers. In cloud computing data center, such technologies as VLAN and distributed virtual switch are adopted to realize isolation between user systems. Distribution intrusion detection and virus protection system are used to resist attack from off-campus network. In addition, other security measures and technologies can be adopted, such as port binding and establishment of virtual firewall.

1.1.3. Data security

Data center is the center of virtualization and efficient management of cloud computing so as to achieve the purpose of centralized management, flexible access and data security. As data are stored in cloud computing platform and security threats faced by them are more prominent. According to data information security standard requirements under the environment of cloud computing, technical 1244 means should be taken in such aspects as data access control, data encryption, data isolation, data residual, data backup and data audit to ensure the security of data in campus cloud.

1) Access control

Different data have different permissions. To maintain the privacy of data, only users with permission can access data. Campus cloud can establish unified identity management, security certificate and access permission control and use authentication technologies such as digital signature, single sign on authentication and double-factor login authentication to ensure data security.

2) Data encryption

Data encryption is an important method to ensure data privacy. Important sensitive data are encrypted automatically. Even if they are filched by illegal users, it is unnecessary to worry about data leakage. Symmetric encryption algorithm with a high encryption performance should be selected for encryption transmission and storage of data. In addition, it is necessary to encryption file system. The robustness of encryption algorithm depends on good key management technology. Meanwhile, key management is the premise of realizing user identity identification and authentication.

3) Data isolation

The establishment of software system of campus cloud widely uses single software system case to serve multiple customers. This means that data of all users are in the sharing environment under the multi-tenant environment. Therefore, it is necessary to

develop an extra data isolation mechanism to ensure invisibility of data among customers. Currently, several mature structures are used to help the system realize data isolation, including Shared Schema Multi-Tenancy, Separated Database and Shared Database Separated Schema. In addition, physical isolation and virtualization can be used to realize security isolation of data and configuration information among different tenants.

4) Data residual

In campus cloud, data residual means that data remain after wiped in certain form. Some physical characteristics might remain after data are wiped so that data can be reconstructed. Therefore, data residual might make some sensitive information leaked unconsciously. Therefore, before reallocation of storage resources, complete removal of data can be ensured with such technologies and methods as destruction of relevant media of enciphered data, destruction of storage medium, disk wiping and content discovery so as to avoid illegal reconstruction of data.

5) Data backup and restoration

When data are stored in campus cloud, users will worry about the loss of data. To ensure absolute security of user data and cope with data loss and business cease caused by emergency and extreme case, campus cloud should be able to implement disaster recovery plan rapidly and continue to provide services. Therefore, it is required to improve disaster recovery backup mechanism of smart campus and improve the robustness of cloud computing system. Data auditing to ensure the accuracy and effectiveness of data, a third-party certification authority is often introduced for data auditing. An independent third-party conduct monitoring and evaluation of business users and provides security auditing report specific to operation and maintenance management of cloud computing supplier. This is essential to ensure cloud computing service.

1.1.4. Service security

Safe smart campus should allow authorized users to use cloud computing service and can still provide effective services for authorized users when the system is damaged partially or it is required to degrade for use. To ensure the demand of the system for availability, cloud computing system can introduce such mechanisms as identification and authentication, access control, data flow control and audit. Due to intensive concentration of users and information resources, it easily becomes the target of hacker attack. Meanwhile, as the consequence and destructive effect caused by denial of service attack will obviously exceed traditional network application environment, we can strengthen the security of host, restrict communication channel with poor foundation and adjust firewall and rules of intrusion detection system to ensure service security.

1.1.5. Application security

Application security 1245 It is required to establish strict cloud computing AAA mechanism through identity authentication and security access control, implement strict identity management, security authentication and access permission control, provide user access records, realize traceability of access and ensure the security of user application.

1.1.6. Management security

Currently, most safety accidents in the network are caused by internal workers of network. Therefore, to realize the security of system structure of smart campus, administrative management is extremely important for the realization of security besides the guarantee of technical security. It is required to establish security management system and security auditing system according to practical situation of the school and record and maintain various security log contents so as to standardize the operation and behaviors of administrative staff and improve post review ability of illegal trace to the source.

1.2. Privacy-preserving Sharing of IoT Data

To the large extent, the IoT data may be of personal nature and therefore it is important to protect it from unauthorized entities accessing it. Privacy is one of the most sensitive subjects in any discussion of IoT protection. Therefore, data privacy is one of the crucial aspects of IoT. The amount of data generated by IoT will be huge. Single pieces of information, i.e., single measurements, in most cases do not represent a significant threat for the owners of IoT devices (temperature at a location, even heart rate of a person at a given moment). However, given that the devices are generating data continuously, it is obvious that unauthorized access to such wealth of data can cause significant problems and can be used to harm the owners of the data (and possibly others, depending on the context of the data). Therefore, it is of paramount importance to protect access to IoT data.

On the other hand, the power of IoT lies in the ability to share data, combine different inputs, process it and create additional value. Hence, it is equally important to enable access to data generated by other IoT devices, while preventing the use of data in un-authorized or undesired ways. The issue of privacy is concerned with authorization and authentication mechanisms. This includes a policy language to define which attributes (roles, identity, etc.) and credentials are requested to grant access to resources. It includes a (data handling) policy language that defines how the requested data (attributes and credentials) is handled and to whom it is passed on. It is also important to consider the mechanisms enabling the protection of information based on encryption algorithms within the secure storage. In terms of the privacy policy implementation, one of the viable solutions is privacy by design, in which users would have the tools they need to manage their own data. The fundamental privacy mechanisms lie in the intelligent data management so that only the required data is collected. Detecting the redundancy, data is anonymized at the earliest possible stage and then deleted at the earliest convenience.

Furthermore, the processing of collected data will have to be minimized according to a strict set of rules so that it cannot be re-used. The proposed approach will define such

methodology together with the mechanisms for the secure storage based on efficient cryptographic algorithms suited for the resource constrained environments. Misconceptions of what "personally identifiable information" is have originated multiple privacy scandals over the last years. Naively anonymized data that rely on the fallacious distinction between "identifying" and "nonidentifying" attributes are vulnerable to re-identification attacks. Notable examples of supposedly anonymized data release which led to lawsuits in the US are the AOL search queries and the Netflix Prize dataset. While some attributes can be identifying by themselves any attribute can be identifying in combination with others. Information disclosure access control must be aware of metrics drawn from data analysis to assess the true risks of privacy breaches. In order to do that, concepts like K-Anonymity and Differential Privacy will be used.

1.3. Threats and Attacks

Threat is a possible danger that might exploit a vulnerability to breach security and therefore cause possible harm. Attack is any attempt to expose, alter, disable, destroy, steal or gain unauthorized access to or make unauthorized use of an Asset.

The IoT is facing various types of attacks including active attacks and passive attacks that may easily disturb the functionality and abolish the benefits of its services. In a passive attack, an intruder just senses the node or may steal the information but it never attacks physically. However, the active attacks disturb the performance physically. These active attacks are classified into two further categories that are internal attacks and external attacks. Such vulnerable attacks can prevent the devices to communicate smartly. Hence the security constraints must be applied to prevent devices from malicious attacks. Different types of attack, nature/behavior of attack and threat level of attacks are discussed in this section. Different levels of attacks are categorized into four types according to their behavior and propose possible solutions to threats/attacks.

1) Low-level attack:

If an attacker tries to attack a network and his attack is not successful.

2) Medium-level attack:

If an attacker/intruder or an eavesdropper is just listening to the medium but don't alter the integrity of data.

3) High-level attack:

If an attack is carried on a network and it alters the integrity of data or modifies the data.

4) Extremely High-level attack:

If an intruder/attacker attacks on a network by gaining unauthorized access and performing an illegal operation, making the network unavailable, sending bulk messages, or jamming network.

The Table given on the next page presents different types of attacks, their threat levels, their nature/behavior, and possible solution to handle these attacks.

Type	Threat level	Behavior	Suggested Solution
Passive	Low	Usually breach data confidentiality. Examples are passive eavesdropping and traffic analysis. Hostile silently listen the communication for his own benefits without altering the data.	Ensure confidentiality of data and do not allow an attacker to fetch information using symmetric encryption techniques.
Man in the Middle	Low to Medium	Alteration and eavesdropping are the examples of this attack. An eavesdropper can silently sense the transmission medium and can modify the data if encryption is not applied and steal the information that is being transmitted. Hostile may also manipulate the data.	Apply data confidentiality and proper integration on data to ensure integrity. Encryption can be also applied so that no one can steal the information or modify the information or encode the information before transmission.
Eavesdropping	Low to Medium	The information content may be lost by an eavesdropper that silently senses the medium. For example, in medical environment, privacy of a patient may be leaked.	Apply encryption on all the devices that perform communication.
Gathering	Medium to High	Occurs when data is gathered from different wireless or wired medium. Examples are skimming, tampering and eavesdropping. Data is being collected to detect messages. Messages may also be altered.	Encryption can be applied to prevent this kind of attack. Identity based method and message authentication code can also be applied in order to prevent the network from such malicious attacks.
Active	High	Effects confidentiality and integrity of data. Hostile can alter the integrity of messages, block messages, or may re-route the messages. It could be an internal attacker.	Ensure both confidentiality and integrity of data. To maintain data confidentiality, symmetric encryption can be applied. An authentication mechanism may be applied to allow data access to only authorized person.
Imitation	High	It impersonates for an unauthorized access. Spoofing and cloning are the examples of this attack. In spoofing attack a malicious node impersonate any other device and launch attacks to steal data or to spread malware. Cloning can re-write or duplicate data.	To avoid from spoofing and cloning attacks, apply identity-based authentication protocols. Physically unclonable function is a countermeasure for cloning attack.

Privacy	High	Sensitive information of an individual or group may be disclosed. Such attacks may be correlated to gathering attack or may cause an imitation attack that can further lead to exposure of privacy.	Apply anonymous data transmission. Transmit sample data instead of actual data. Can also apply techniques like ring signature and blind signature.
Interruption	High	Affects availability of data. This makes the network unavailable.	Applying authorization, only authorized users are allowed to access specific information to perform certain operation.
Routing Diversion	High	Only the route is diverted showing the huge traffic and the response time increased.	Ensure connectivity based approach so no route will be diverted.
Blocking	Extremely High	It is type of DoS, jamming, or malware attacks. It sends huge streams of data which may leads to jamming of network, similarly different types of viruses like Trojan horses, worms, and other programs can disturb the network.	Turn on the firewall, apply packet filtering, anti-jamming, active jamming, and updated antivirus programs in order to protect the network from such attacks.
Fabrication	Extremely High	Affects the authenticity of information. Hostile can inject false data and can destroy the authenticity of information.	Data authenticity can be applied to ensure that no information is changed during the transmission of data.
DoS	Extremely High	Malicious user may modify the packets or resend a packet again and again on network. User can also send bulk messages to devices in order to disturb the normal functionalities of devices.	Apply cryptographic techniques to ensure security of network. Apply authenticity to detect the malicious user and block them permanently. In this way, the network is prevented from damage.

So, according to our system design which is basically IoT Base Smart Campus System there will a Index base queries Handling for all user.

> **"A system which communicate through Protocol but simply indexed to an IP is very Dangerous"**
>
> ***Shahid***

so, there can be attack of various kind like Following

- **ARP Spoofing**
- **ARP Sniffing**
- **Netcut**
- **DNS Spoofing**
- **Download Replace**
- **Code Injector**
- **Bypassing HTTPS**

So, we will carry on to These attacks all detail and method is explained in respective attack
Language for script : Python 3, Java Script

NOTE:

All These Attacks Scripts will work on Our IoT Base Smart Campus as well as all HTTP/HTTPS site so We do not recommend to Run This Script of Any Public/Private Site As we know it's Illegal According to Cyber Law Islamic Republic of PAKISTAN**.** It is only Tested On conceptualizing Smart Campus System Index

ARP Spoofing

Concept

The Address Resolution Protocol (ARP) Spoofing attack, also called ARP Cache Poisoning or ARP Poison Routing, is a technique by which an attacker sends spoofed ARP messages onto a Local Area Network (LAN). It is used to allow the attacker access to incoming internet traffic on a LAN by having their Media Access Control (MAC) Address be linked to the Internet Protocol (IP) Address of another host (usually, the default gateway). Through this, they're able to receive incoming traffic intended for that IP Address which allows them to intercept the data, modify traffic, or even stop all traffic on the network. Because of this, the technique is often used to open up the possibility of other attacks such as a Denial of Service (DoS) attack, a man in the middle attack, and a session hijacking attack. The success of the attack depends heavily on the attacker gaining direct success to the targeted local network segment and it can only be used on networks which use ARP.

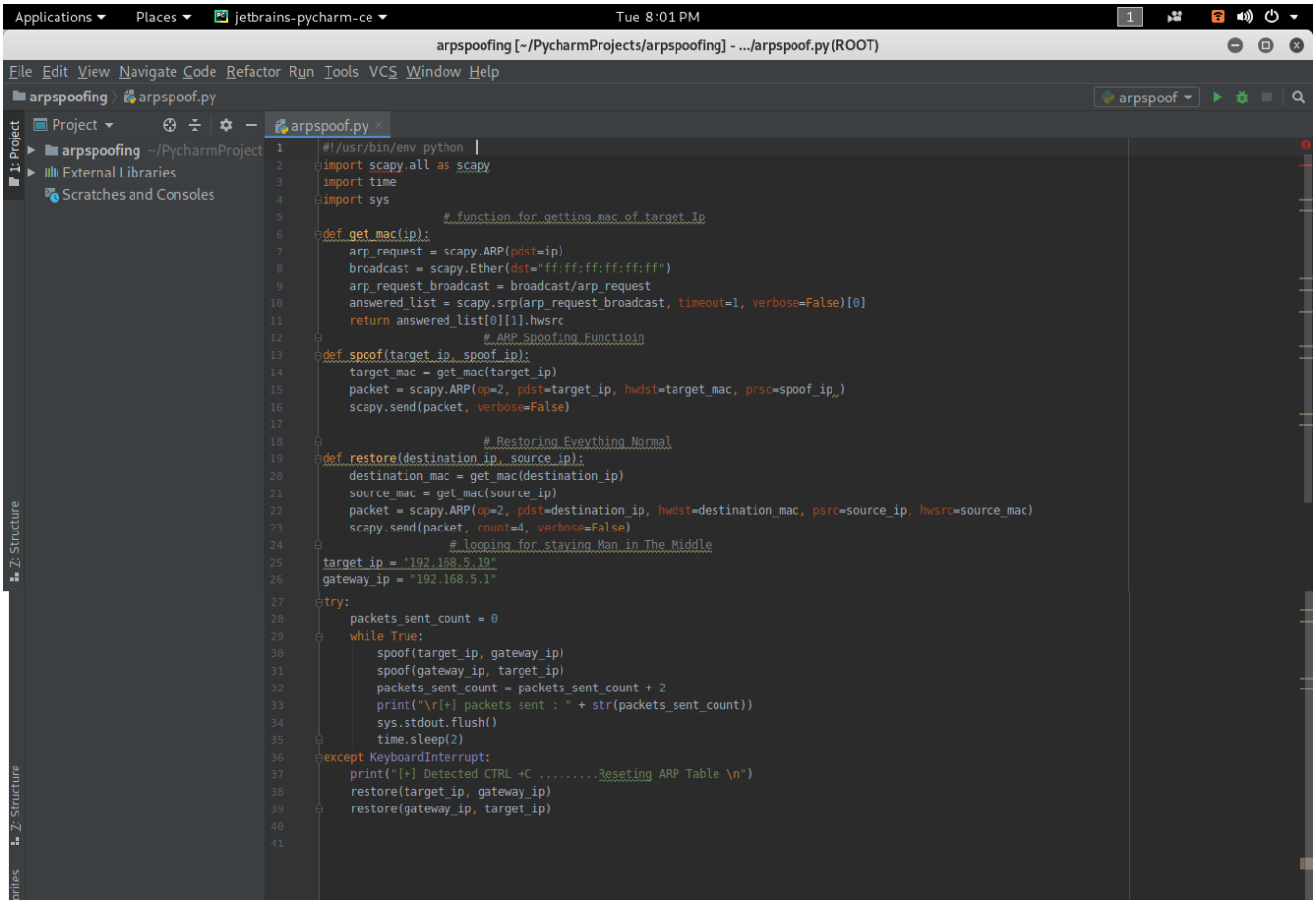
ARP Vulnerabilities

The ARP is a common protocol used for communication to resolve Internet layer addresses into link layer addresses. When a datagram is sent from one device to another on a LAN, the destination Internet Protocol (IP) Address must be changed to a MAC Address for transmission across the data link layer. This is accomplished by a device, whose IP Address is known to the network, sending out a broadcast packet (an ARP Request) to all devices on the local network. The intended recipient of the datagram with the matching IP Address in the request responds to the broadcast with an ARP Reply which contains the MAC Address for the IP.

The protocol is stateless and network hosts automatically cache all ARP Replies they receive regardless of whether network hosts requested them or not. Even replies which have yet to expire will be overwritten by this behavior. As there is no method in the protocol to help authenticate which peer the packet originated from, it is possible to take advantage of the protocol through ARP Spoofing.

So, this will be very Helpful as an attacker to launch a attack on IoT Base Smart Campus System to get Some Knowledge Which Student/Faculty member doing what in Smart System.

SCRIPT



Requirement:

- Kali Linux
- Pycharm with Scapy and pip3 installed

Steps:

- Discover your Target IP by using Command

```
$ netdiscover -r <your IP with /Range>
```

- Then put your router and target IP in script
- Then run this script by following commands

```
$ python arpspoof.py
```

Then this for queue Creation and Port Forwarding

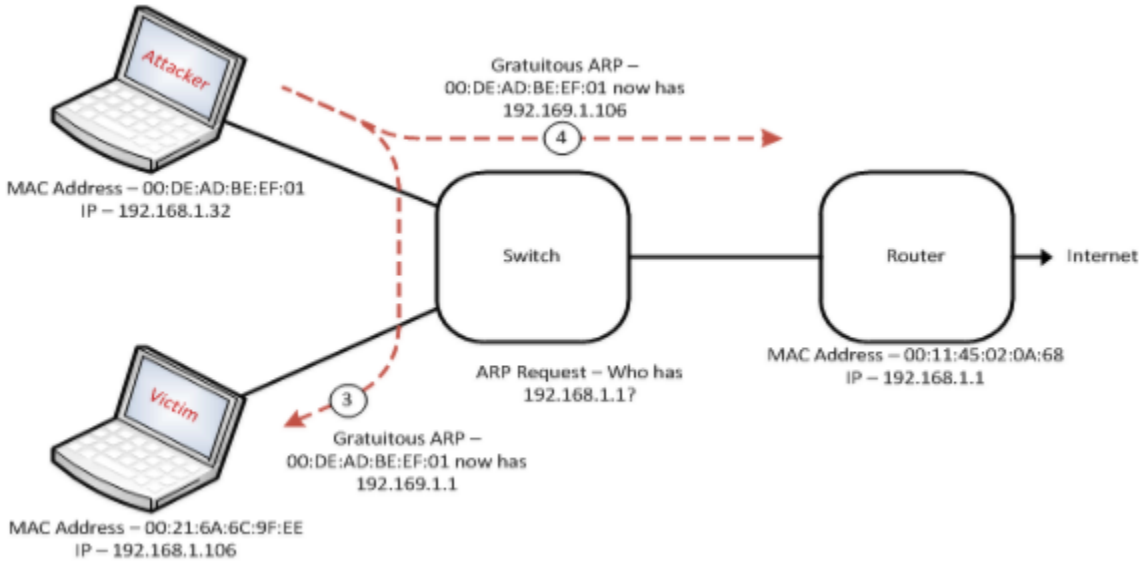
```
$iptables -I FORWARD -j NFQUEUE --queue-num 0
```

```
$ echo 1 > /proc/sys/net/ipv4/ip_forward
```

ARP Spoofing

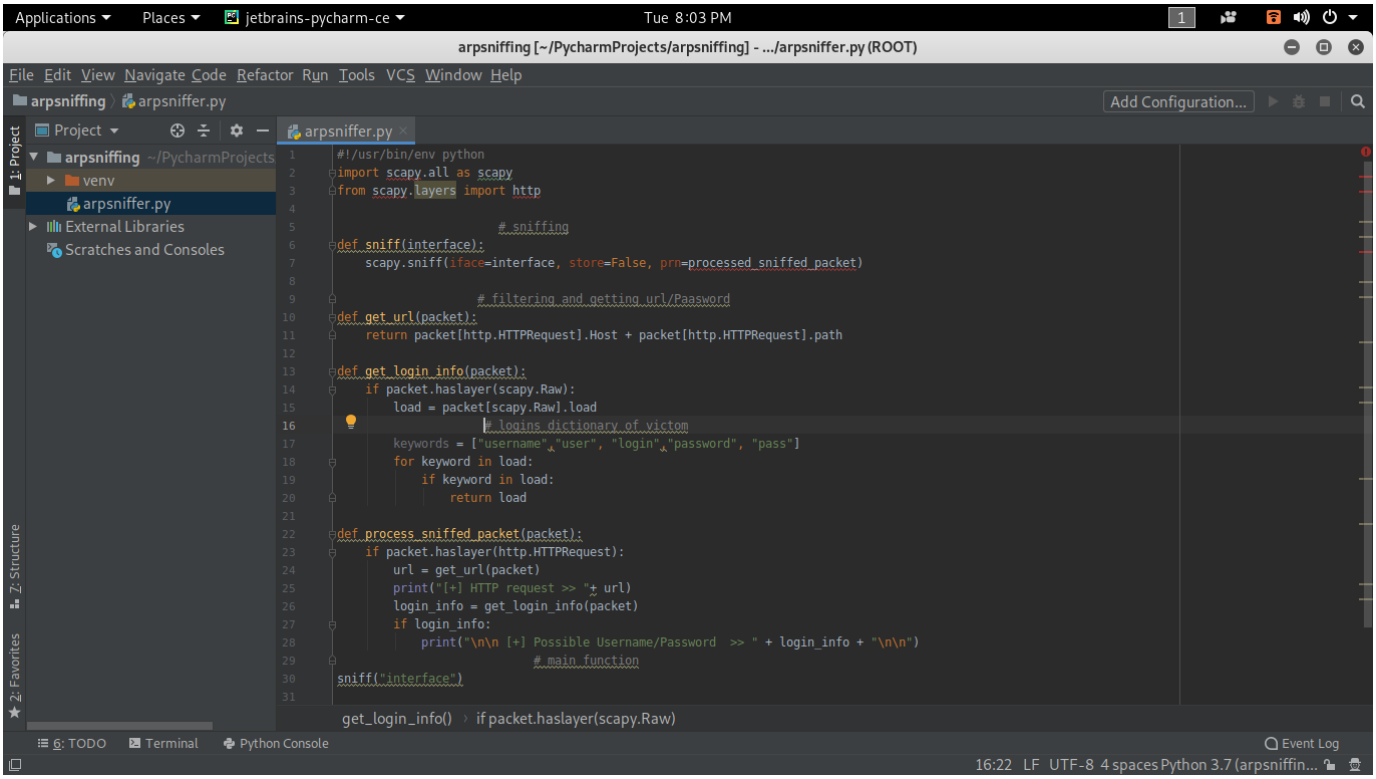
It's a targeted attack, where attacker sniffs the traffic and chooses targets. It uses ARP poisoning method to forge fake ARP request/reply to targets forcing them to update their ARP cache with MAC address of Attacker machine in the place of the genuine target. So the traffic between target's will be split into two. One connection between target1 and attacker and other between target2 and attacker. So the attacker being the man in the middle can modify/replay the traffic. He will be able to capture sensitive information between the targets.

In this case, an attacker can spoof the MAC address of a trusted host and forge ARP request/replies to overload the Switch. Then the switch is set in "forward mode" an attacker can now sniff the packets on the traffic.



So This will be Very Helpful so Sniff Package we can do it with EtterCap but this script does it for all

SCRIPT



Requirement:

- Kali Linux
- Pycharm with Scapy and pip3 installed

Steps:

- Run Arp spoofing attack
- Then run this script by following commands

```
$ python arpsniffier.py
```

Then Do not Forget queue Creation and Port Forwarding

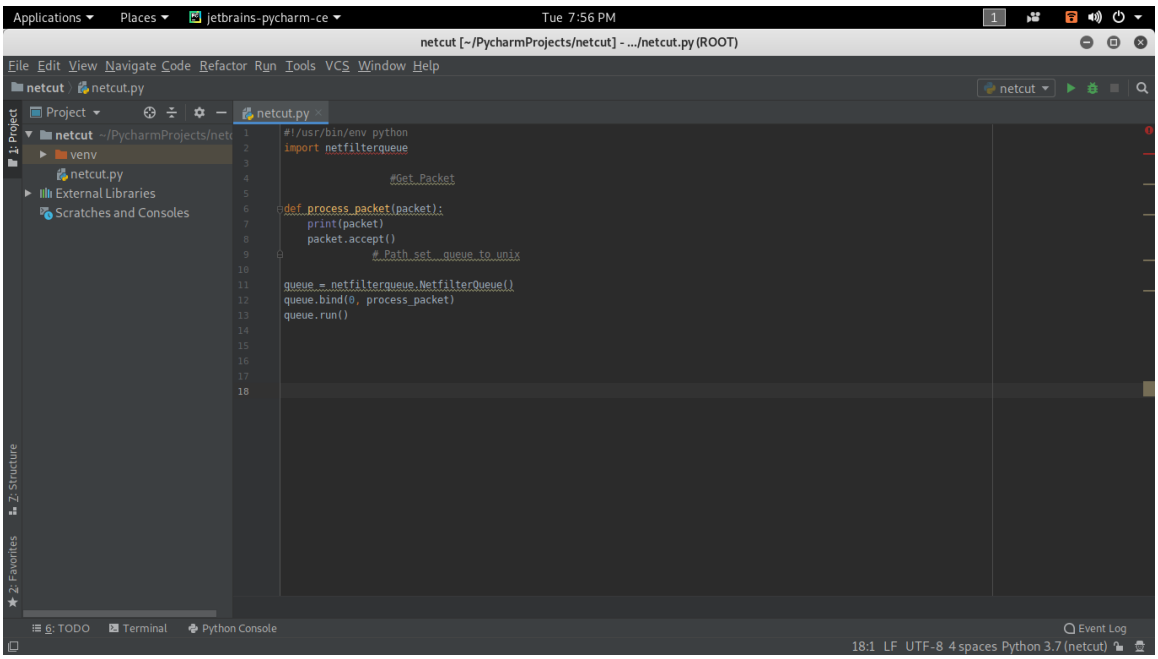
```
$iptables -I FORWARD -j NFQUEUE --queue-num 0
```

```
$ echo 1 > /proc/sys/net/ipv4/ip_forward
```

NETCUT

This Attack will suspend services like smart Library etc

SCRIPT



Requirement:

- Kali Linux
- Pycharm with Scapy and pip3 installed

Steps:

Run queue Creation and Port Forwarding

```
$iptables -I FORWARD -j NFQUEUE --queue-num 0
```

```
$ echo 1 > /proc/sys/net/ipv4/ip_forward
```

Then run this script by following commands

```
$ python netcut.py
```

DNS Spoofing

(This is for whole Network)

Domain Name Server (DNS) spoofing (a.k.a. DNS cache poisoning) is an attack in which altered DNS records are used to redirect online traffic to a fraudulent website that resembles its intended destination.

Once there, users are prompted to login into (what they believe to be) their account, giving the perpetrator the opportunity to steal their access credentials and other types of sensitive information. Furthermore, the malicious website is often used to install worms or viruses on a user's computer, giving the perpetrator long-term access to it and the data it stores.

Methods for executing a DNS spoofing attack include:

Man in the middle (MITM) – The interception of communications between users and a DNS server in order to route users to a different/malicious IP address.

DNS server compromise – The direct hijacking of a DNS server, which is configured to return a malicious IP address.

DNS cache poisoning example

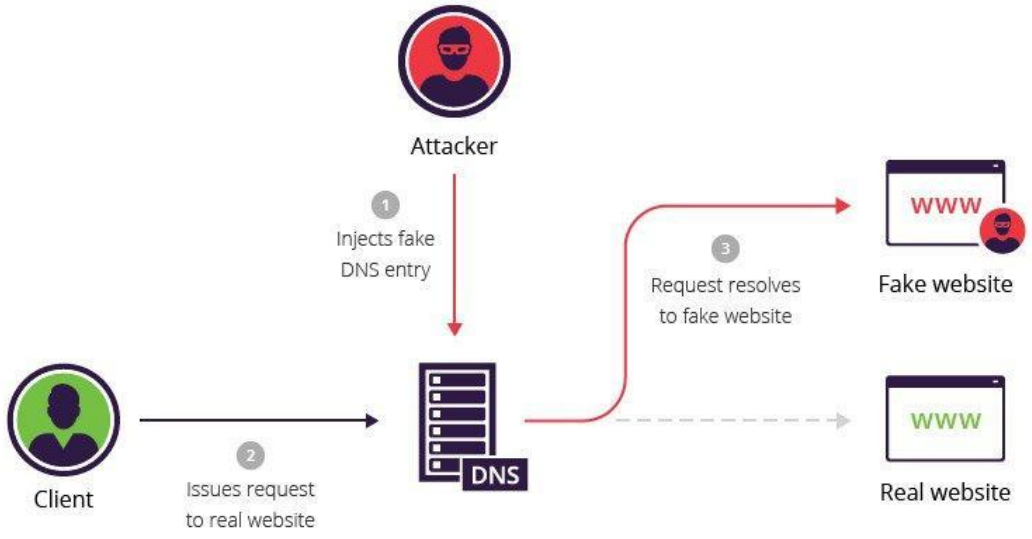
The following example illustrates a DNS cache poisoning attack, in which an attacker (IP 192.168.3.300) intercepts a communication channel between a client (IP 192.168.1.100) and a server computer belonging to the website www.smartcampus.com (IP 192.168.2.200).

In this scenario, a tool (e.g., ARP Spoof) is used to dupe the client into thinking that the server IP is 192.168.3.300. At the same time, the server is made to think that the client's IP is also 192.168.3.300.

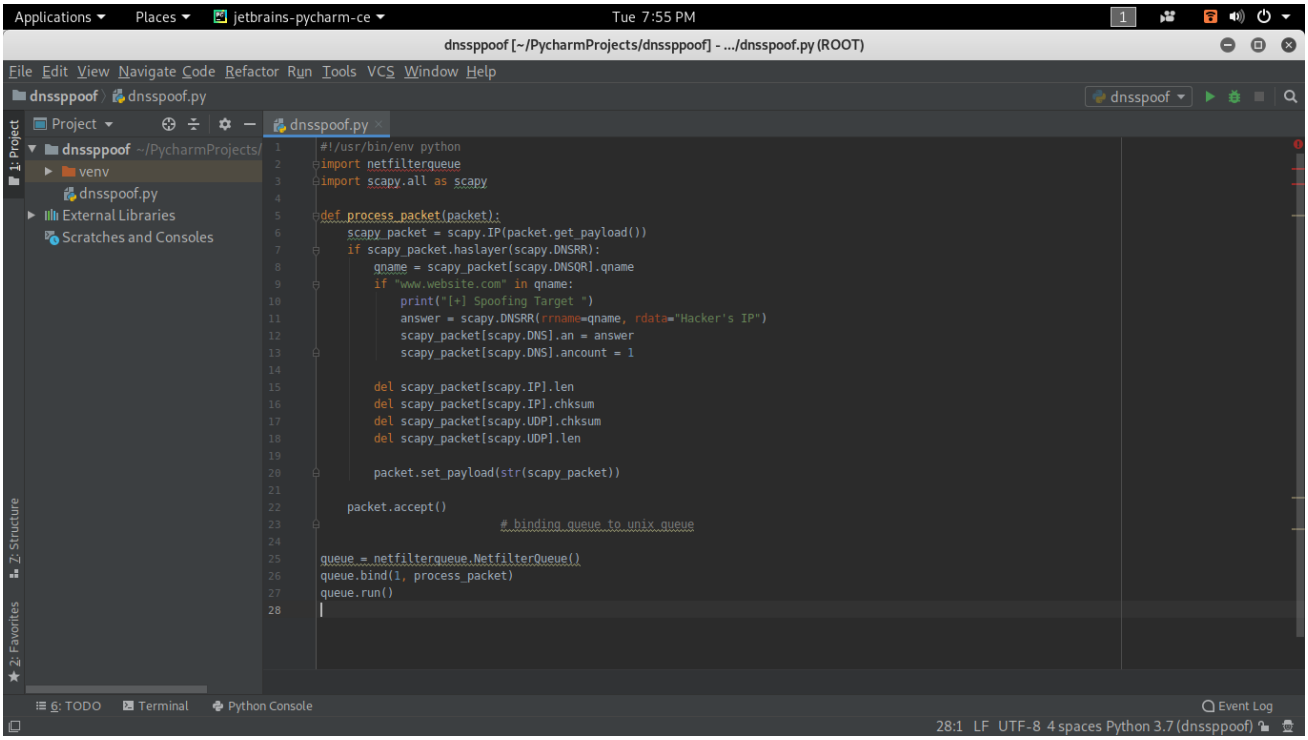
Such a scenario would proceed as follows:

- The attacker uses ARP Spoof to issue the command: ARP Spoof 192.168.1.100 192.168.2.200. This modifies the MAC addresses in the server's ARP table, causing it to think that the attacker's computer belongs to the client.
- The attacker once again uses ARP Spood to issue the command: ARP Spoof 192.168.2.200 192.168.1.100, which tells the client that the perpetrator's computer is the server.
- The attacker issues the Linux command: **echo 1> /proc/sys/net/ipv4/ip_forward**. As a result, IP packets sent between the client and server are forwarded to the perpetrator's computer.

- The host file, 192.168.3.300 estores.com is created on the attacker's local computer, which maps the website www.estores.com to their local IP.
- The perpetrator sets up a web server on the local computer's IP and creates a fake website made to resemble www.estores.com.
- Finally, a tool (e.g., DNS Spoof) is used to direct all DNS requests to the perpetrator's local host file. The fake website is displayed to users as a result and, only by interacting with the site, malware is installed on their computers.



SCRIPT



Steps:

- Run Queue Creation and Port forward
- Run NETCUT script
- Run This Script

DOWNLOAD REPLACE

Once we able to exploit DNS Spoofing The we will be able to do many things like

- We replace Fake time Table is Smart Campus system
- Fake result uploaded recently
- Fake book equipped from Library ETC

Remember, even though this hack requires that you be on the same LAN, you could get access to the LAN through a remote vulnerability or a weak password on just ONE machine on the network. In institutions with thousands of computers on their network, that means you must find a single machine that is exploitable to be able implement this attack for the entire network.

DNS Spoofing: Redirecting Users to Your Website

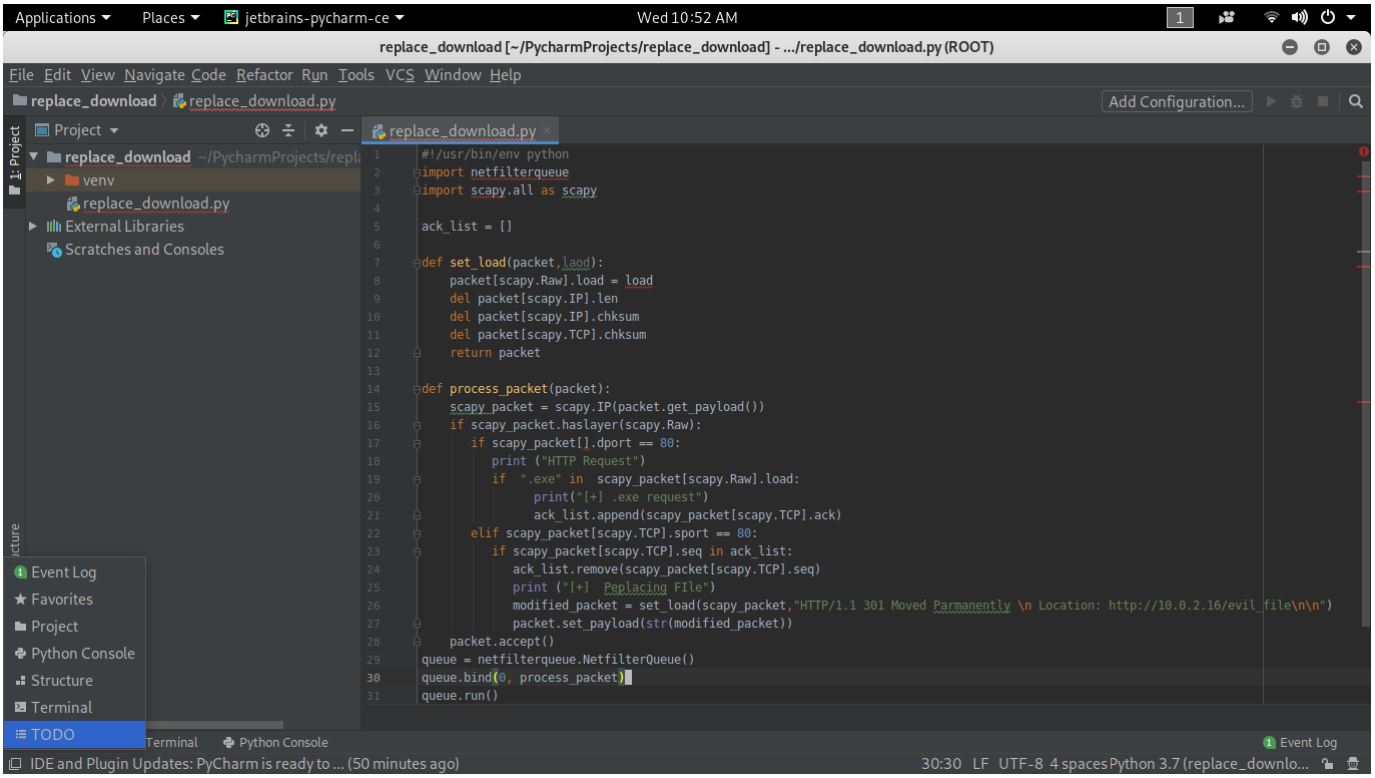
In this hack, we will be exploiting the Domain Name Service (DNS). As you know, DNS is used for domain name resolution or converting a domain name such as download/notes/smartcampus.com to an IP address, 192.168.65.101. If we can mess with this protocol, we could very well send someone looking for a domain name such as bankofamerica.com to our malicious website and harvest their credentials.

Pre Script-Precaution

- Make Sure you know what file you should replace
- Make sure you have EVIL file which is To be replace

NOW FIRE UP KALI >>

SCRIPT:



Requirement:

- Kali Linux
- Pycharm with Scapy and pip3 installed

Steps:

- Run queue Creation and Port Forwarding

```
$iptables -I FORWARD -j NFQUEUE --queue-num 0
```

```
$ echo 1 > /proc/sys/net/ipv4/ip_forward
```

- Then Run DNS Spoofing

```
$ python dnsspoof.py
```

Then Run This Script Using

```
$Python download_replace.py
```

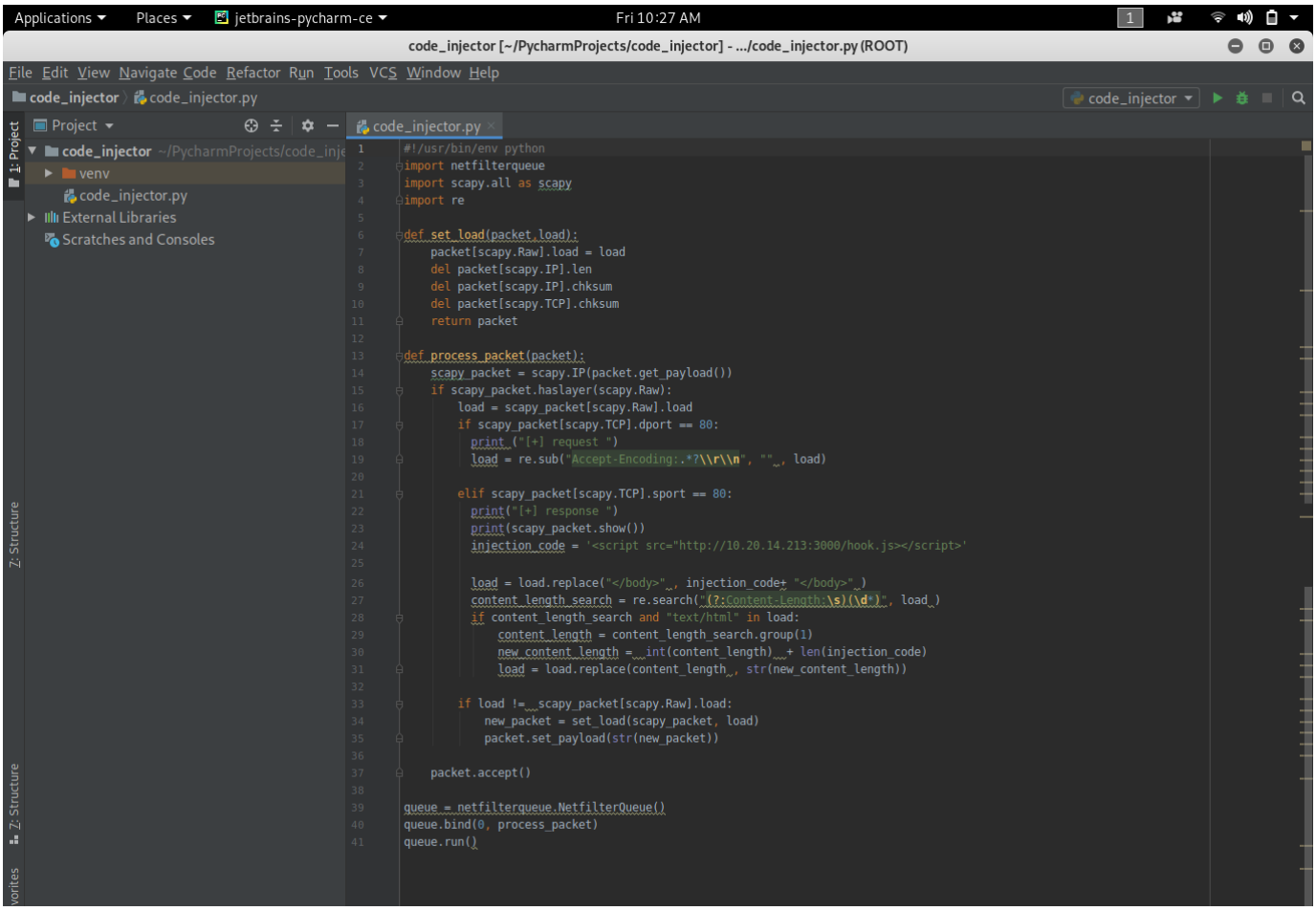
CODE_INJECTOR

As We are in the network So we can execute any Java Script in Targeted System

- We can return cache Return
- We can get password from Raw Layer
- We can show Alert “You are Hacked”

Or we can execute any script to inject malware of any kind after injecting we can create session and remotely control targeted victim.

SCRIPT:



STEP:

- first run arp spoofing
- Then run this script
- And open BeEFx on kali to Hook Victim

Bypassing HTTPS

If you require to attack on HTTPS index then this script will work fine
As we know IoT is working on simple FTP or HTTP Protocol so we did not need This But
We will do it

So, for following both attack

1. DOWNLOAD_REPLACE HTTTPS
2. CODE_INJECTOR HTTPS

We Designed two Scrip but first we will provide you steps and Intro of SSLStrip

SSLStrip:

sslstrip is a tool that transparently hijacks HTTP traffic on a network, watch for HTTPS links and redirects, and then map those links into look-alike HTTP links or homograph-similar HTTPS links. It also supports modes for supplying a favicon which looks like a lock icon, selective logging, and session denial.

STEPS:

- Run ARP spoof script
- Run SSLStrip by typing SSLStrip in Terminal
- Commands of port redirection 10000 (SSLStrip)
- Then desire attack but if Download_Replace attack ten change iptables rule to your kali machine then port forward to SSLStrip.

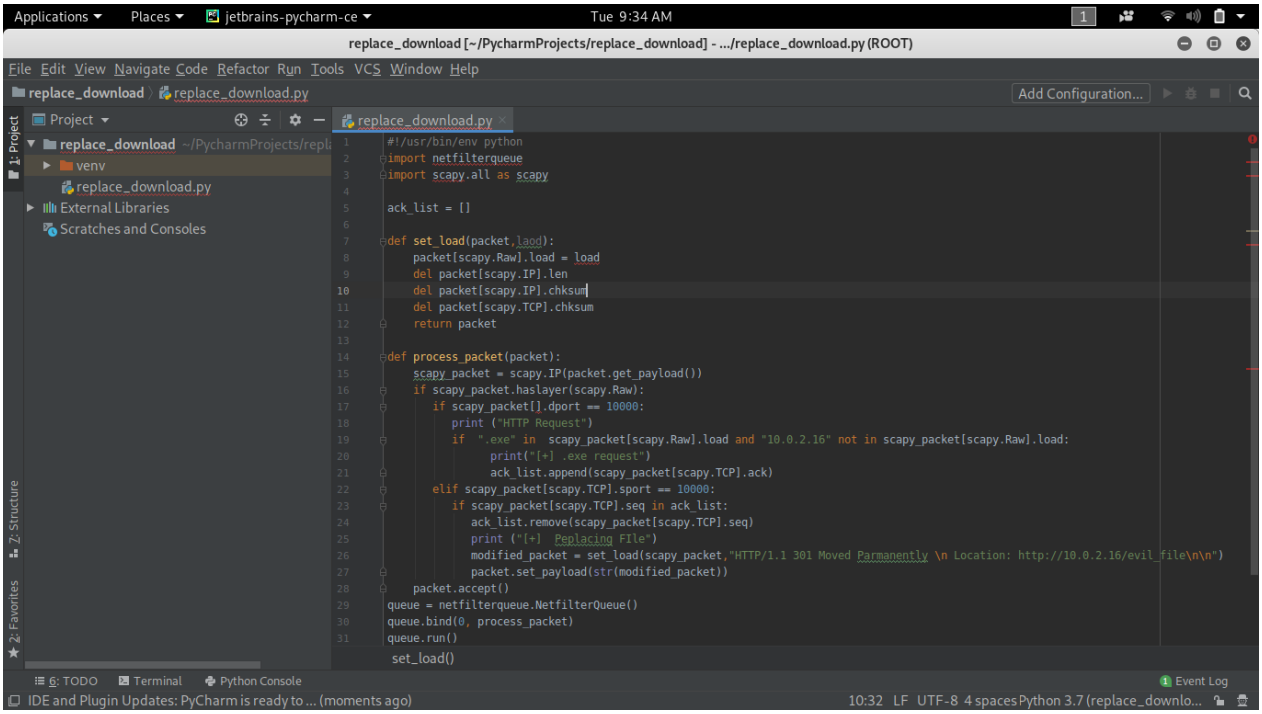
PORT Forwarding in SSLStrip:

- For Port Forwarding in SSLStrip Particularly we use This Commands

$ iptabels -t nat -A PREROUTING -p tcp --destination-port 80 -j REDIRECT --to-port 10000

Then Run Any Attack You Want from Both for HTTPS sites

SCRIPT for Download_Replace in HTTPS:



SCRIPT for Code_Injector in HTTPS:

